

List of Key SOTA similarity measures in collaborative filtering

The following table lists key SOTA measures, while the extensions are detailed in the next sections.

Measure	Group	Equation
AU01.Jaccard	1.Jaccard	(1.1)
AU02.Cosine	2.Cosine	(2.1)
AU03.Pearson	3.Pearson	(3.1)
AU04.MSD	4.MSD	(4.1)
AU05.SRC	5.SRC	(5.1)
AU06.PIP	6.PIP	(6.1)
AU07.PSS	7.PSS	(7.1)
AU08.BCF	8.BCF	(8.1)
AU09.MMD	9.MMD	(9.1)
AU10.Triangle	10.Triangle	(10.1)
AU11.Feng	11.Feng	(11.1)
AU12.Mu	12.Mu	(12.1)
AU14.SMD	14.SMD	(14.1)
AU15.NNSM	15.NNSM	(15.1)
AU16.TA	16.TA	(16.1)
AU17.RA	17.RA	(17.1)
AU18.Entropy	18.Entropy	(18.1)
AU19.KL	19.KL	(19.1)
AU26.URP	26.URP	(26.1)
AU27.STB	27.STB	(27.1)
AU30.Singularity	30.Singularity	(30.1)
BU01.Dice	1.Jaccard	(1.2)
BU01.IJ	1.Jaccard	(1.4)
BU01.JacDual	1.Jaccard	(1.14)
BU01.JacEDS	1.Jaccard	(1.13)
BU01.JacLMH	1.Jaccard	(1.6)
BU01.JacRA	1.Jaccard	(1.11)
BU01.JaccardMulti	1.JaccardMulti	(1.3)
BU01.RatingJaccard	1.Jaccard	(1.8)
BU01.RelevantJaccard	1.Jaccard	(1.10)
BU02.CosineJ	2.Cosine	(2.5)
BU03.CPC	3.Pearson	(3.2)
BU03.IPC	3.Pearson	(3.5)
BU03.IPWR	3.Pearson	(3.6)
BU03.PC	3.Pearson	(3.8)
BU03.PearsonJ	3.Pearson	(3.7)
BU03.SPC	3.Pearson	(3.4)
BU03.WPC	3.Pearson	(3.3)
BU04.MSDJ	4.MSD	(4.3)
BU06.MPIP	6.PIP	(6.8)
BU07.NHMS	7.PSS	(7.4)
BU07.NHMS.Amer	7.PSS	(B7.2)
BU07.NHMS.SMD	7.PSS	(B7.1)
BU07.PSSJ	7.PSS	(7.3)

BU08.BCFJ	8.BCF	(8.3)
BU09.CjacMD	9.MMD	(9.2)
BU10.TMJ	10.Triangle	(10.2)
BU14.Amer	14.SMD	(14.4)
BU14.AmerJ	14.SMD	(B14.2)
BU14.HSMD	14.SMD	(14.2)
BU14.HSMDJ	14.SMD	(14.3)
BU14.QTI	14.SMD	(14.9)
BU14.QTIJ	14.SMD	(14.10)
BU14.SMDJ	14.SMD	(B14.1)
BU15.NNSMJ	15.NNSM	(15.2)
BU16.TAJ	16.TA	(16.2)
BU16.TAN	16.TA	(16.3)
BU16.TANJ	16.TA	(16.4)
CU01.IJ.Amer	1.Jaccard	(C1.4)
CU01.IJ.Cosine	1.Jaccard	(C1.1)
CU01.IJ.PSS	1.Jaccard	(C1.3)
CU01.IJ.Pearson	1.Jaccard	(C1.2)
CU01.IJ.TA	1.Jaccard	(C1.5)
CU14.Amer.CPC	14.SMD	(C14.3)
CU14.Amer.Cosine	14.SMD	(C14.1)
CU14.Amer.NNSM	14.SMD	(C14.6)
CU14.Amer.PSS	14.SMD	(C14.4)
CU14.Amer.Pearson	14.SMD	(C14.2)
CU14.Amer.QTI	14.SMD	(C14.5)
CU14.Amer.TA	14.SMD	(C14.7)

Table 0.3. Sample conventional measures

Note, equations B^* and C^* are listed in appendices B and C .

1. Jaccard

The first measure which is described here is Jaccard because of its special feature when it does not concern magnitude of numeric rating values. Jaccard measure is ratio of cardinality of common set $I_1 \cap I_2$ to cardinality of union set $I_1 \cup I_2$. It measures how much common items both users rated, which is defined as follows (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 158):

$$\text{Jaccard}(u_1, u_2) = \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} \quad (1.1)$$

A simple version of Jaccard is Dice measure as follows:

$$\text{Dice}(u_1, u_2) = \frac{2|I_1 \cap I_2|}{|I_1 \cup I_2|} \quad (1.2)$$

Although Jaccard is simple, it is effective within sparse rating matrix which has many missing values. Another version of Jaccard is (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 158):

$$\text{JaccardMulti}(u_1, u_2) = \frac{|I_1 \cap I_2|}{|I_1||I_2|} \quad (1.3)$$

Liang, Ma, and Yuan (Liang, Ma, & Yuan, 2015) proposed an improved variant of Jaccard based on the concept of singularity. Let r_m be median of rating values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$, hence, rating values which are greater or equal than r_m are called relevant ratings and the others which are smaller than r_m are called

non-relevant. Liang et al. called “relevant” and “non-relevant” as “positive” and “negative”, respectively. Given user 1 and user 2, for every item j , let:

- If both r_{1j} and r_{2j} are positive, item j belongs to the predefined set PA .
- If both r_{1j} and r_{2j} are negative, item j belongs to the predefined set NA .
- If r_{1j} and r_{2j} are different which means that one is positive and the other is negative, item j belongs to the predefined set D .
- When one user rated item j but the other did not rate, if such rating is positive then item j belongs to the predefined set PO , otherwise item j belongs to the predefined set NO .

Liang et al. (Liang, Ma, & Yuan, 2015) defined singularity quantities such as positive singularity, negative singularity, and empty singularity denoted S_j^P , S_j^N , and S_j^E as follows:

$$S_j^P = 1 - \frac{|P_j|}{|U|}$$

$$S_j^N = 1 - \frac{|N_j|}{|U|}$$

$$S_j^E = 1 - \frac{|E_j|}{|U|}$$

Where P_j is the set of users who gave positive ratings to item j , N_j is the set of users who gave negative ratings to item j , and E_j is the set of users who did not rate item j . Liang et al. defined Improved Jaccard (IJ) as follows (Liang, Ma, & Yuan, 2015, p. 1634):

$$IJ(u_1, u_2) = \left(\sum_{j \in PA} S_j^P + \sum_{j \in NA} S_j^N + \sum_{j \in D} \sqrt{S_j^P S_j^N} \right) / \left(\sum_{j \in PA} S_j^P + \sum_{j \in NA} S_j^N + \sum_{j \in D} \sqrt{S_j^P S_j^N} + \sum_{j \in PO} \sqrt{S_j^P S_j^E} + \sum_{j \in NO} \sqrt{S_j^N S_j^E} \right) \quad (1.4)$$

As a result, when Liang et al. concerned singularity quantities, they concerned specific features of users so that the recommendation process is more appropriate. For example, given two rating vectors $u_3 = (r_{31}=4, r_{32}=1, r_{33}=5, r_{34}=5)$, and $u_4 = (r_{41}=1, r_{42}=2, r_{43}=?, r_{44}=?)$ from table 0.1:

	Item 1	Item 2	Item 3	Item 4
User 1	$r_{11} = 1$	$r_{12} = 2$	$r_{13} = 1$	$r_{14} = 5$
User 2	$r_{21} = 2$	$r_{22} = 1$	$r_{23} = 2$	$r_{24} = 4$
User 3	$r_{31} = 4$	$r_{32} = 1$	$r_{33} = 5$	$r_{34} = 5$
User 4	$r_{41} = 1$	$r_{42} = 2$	$r_{43} = ?$	$r_{44} = ?$

we have $PA = \{\}$, $NA = \{1, 2\}$, $D = \{\}$, $PO = \{3, 4\}$, $NO = \{\}$, $P_1=1$, $N_1=3$, $E_1=0$, $P_2=0$, $N_2=4$, $E_2=0$, $P_3=1$, $N_3=2$, $E_3=1$, $P_4=3$, $N_4=0$, and $E_4=1$ when $r_m=3$. We also obtain $S_1^P = 1 - 1/4 = 3/4$, $S_1^N = 1 - 3/4 = 1/4$, $S_1^E = 1 - 0/4 = 1$, $S_2^P = 1 - 0/4 = 1$, $S_2^N = 1 - 4/4 = 0$, $S_2^E = 1 - 0/4 = 1$, $S_3^P = 1 - 1/4 = 3/4$, $S_3^N = 1 - 2/4 = 1/2$, $S_3^E = 1 - 1/4 = 3/4$, $S_4^P = 1 - 3/4 = 1/4$, $S_4^N = 1 - 0/4 = 1$, and $S_4^E = 1 - 1/4 = 3/4$. As a result, the IJ measure of u_3 and u_4 is:

$$IJ(u_3, u_4) = \frac{S_1^N + S_2^N}{S_1^N + S_2^N + \sqrt{S_3^P S_3^E} + \sqrt{S_4^P S_4^E}} = \frac{1/4 + 0}{1/4 + 0 + \sqrt{3/4 * 3/4} + \sqrt{1/4 * 3/4}} \cong 0.1745$$

Lee (Lee, 2017) divided rating range into three sub-intervals, for each sub-interval a sub-Jaccard measure is computed. Finally, the whole Jaccard measure of Lee called JacLMH is the average of such three sub-Jaccard measures. Let r_L and r_H be minimum value and maximum

value of rating values, for example, if rating values range from 1 to 5 then, $r_L = 1$ and $r_H = 5$. Let L_{bd} and H_{bd} be boundaries of the three sub-intervals such that $r_L < L_{bd} < H_{bd} < r_H$, the set of items rated by user i denoted I_i is divided into three sub-sets I_{Li} , I_{Mi} , and I_{Hi} as follows (Lee, 2017, p. 803):

$$\begin{aligned} I_{Li} &= \{j \in I_i: r_{ij} \leq L_{bd}\} \\ I_{Mi} &= \{j \in I_i: L_{bd} < r_{ij} < H_{bd}\} \\ I_{Hi} &= \{j \in I_i: r_{ij} \geq H_{bd}\} \end{aligned}$$

The three sub-Jaccard measures with regard to such three sub-sets are defined as follows (Lee, 2017, p. 803):

$$\begin{aligned} Jac_L(u_1, u_2) &= \frac{|I_{L1} \cap I_{L2}|}{|I_{L1} \cup I_{L2}|} \\ Jac_M(u_1, u_2) &= \frac{|I_{M1} \cap I_{M2}|}{|I_{M1} \cup I_{M2}|} \\ Jac_H(u_1, u_2) &= \frac{|I_{H1} \cap I_{H2}|}{|I_{H1} \cup I_{H2}|} \end{aligned} \quad (1.5)$$

The final JacLMH is average of such three sub-Jaccard measures (Lee, 2017, p. 803):

$$JacLMH = \frac{1}{3} (Jac_L(u_1, u_2) + Jac_M(u_1, u_2) + Jac_H(u_1, u_2)) \quad (1.6)$$

Traditional Jaccard does not concern numerical rating values and so, Ayub et al. (Ayub, Ghazanfar, Khan, & Saleem, 2020) improved Jaccard measure by numerical equality. Let $N_T(u_1, u_2)$ be the count of ratings that are equal in numerical values, as follows (Ayub, Ghazanfar, Khan, & Saleem, 2020, p. 10003):

$$N_T(u_1, u_2) = \begin{cases} |\{j \in I_1 \cap I_2: r_{1j} = r_{2j}\}| + 1 & \text{if } \forall j \in I_1 \cap I_2, r_{1j} = r_{2j} \\ |\{j \in I_1 \cap I_2: r_{1j} = r_{2j}\}| & \text{otherwise} \end{cases} \quad (1.7)$$

Their proposed measure called RatingJaccard is defined as follows (Ayub, Ghazanfar, Khan, & Saleem, 2020, p. 10003):

$$RatingJaccard(u_1, u_2) = \frac{N_T(u_1, u_2)}{|I_1 \cup I_2|} \quad (1.8)$$

Ayub et al. also improved the quantity $N_T(u_1, u_2)$ by associating it with a pre-defined threshold TH as follows (Ayub, Ghazanfar, Khan, & Saleem, 2020, p. 10003):

$$N_T(u_1, u_2) = \begin{cases} |\{j \in I_1 \cap I_2: r_{1j} = r_{2j}\}| + 1 & \text{if } \forall j \in I_1 \cap I_2, r_{1j} = r_{2j} \text{ or } |\bar{u}_1 - \bar{u}_2| \leq TH \\ |\{j \in I_1 \cap I_2: r_{1j} = r_{2j}\}| & \text{otherwise} \end{cases} \quad (1.9)$$

Where $\bar{u}_1$ and $\bar{u}_2$ are mean values of u_1 and u_2 , respectively.

$$\bar{u}_1 = \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \bar{u}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j}$$

Bag et al. (Bag, Kumar, & Tiwari, 2019) improved Jaccard by concerning more un-co-rated items which are items rated only one user among two concerned users. According to their viewpoint, the traditional Jaccard is re-formulated as follows (Bag, Kumar, & Tiwari, 2019, p. 57):

$$Jaccard(u_1, u_2) = \frac{|I_1 \cap I_2|}{|I_1| + |I_2| - |I_1 \cap I_2|}$$

They defined $|\bar{I}_1|$ and $|\bar{I}_2|$ as the numbers of items which are not rated by only user 1 and only user 2, respectively (Bag, Kumar, & Tiwari, 2019, p. 57).

$$|\bar{I}_1| = |I_1| - |I_1 \cap I_2|, |\bar{I}_2| = |I_2| - |I_1 \cap I_2|$$

The traditional Jaccard is re-written as follows (Bag, Kumar, & Tiwari, 2019, p. 58):

$$\text{Jaccard}(u_1, u_2) = \frac{|I_1 \cap I_2|}{|\bar{I}_1| + |\bar{I}_2| + |I_1 \cap I_2|} = \frac{1}{1 + \frac{|\bar{I}_1| + |\bar{I}_2|}{|I_1 \cap I_2|}}$$

Therefore, Jaccard measure is proportional to two quantities $(|\bar{I}_1| + |\bar{I}_2|)^{-1}$ and $|I_1 \cap I_2|$. Bag et al. (Bag, Kumar, & Tiwari, 2019) improved such proportion by specify that their Jaccard will be proportional to three quantities $|\bar{I}_1|^{-1}$, $|\bar{I}_2|^{-1}$, and $|I_1 \cap I_2|$. As a result, Bag et al defined the so-called relevant Jaccard as follows (Bag, Kumar, & Tiwari, 2019, p. 59):

$$\text{RelevantJaccard}(u_1, u_2) = \frac{1}{1 + \frac{1}{|I_1 \cap I_2|} + \frac{|\bar{I}_1|}{1 + |\bar{I}_1|} + \frac{1}{1 + |\bar{I}_2|}} \quad (1.10)$$

Wu, Huang, and Wang (Jindal, Sharma, & Verma, 2022, p. 4) also improved Jaccard measure by concerning numerical values. Hence, they proposed the so-called JacRA as follows:

$$\text{JacRA}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} \frac{\min(r_{1j}, r_{2j})}{\max(r_{1j}, r_{2j})}}{|I_1 \cup I_2|} \quad (1.11)$$

Percentage of Non-Common Ratings (PNCR) is defined as follows (Gazdar & Hidri, 2019, p. 19):

$$\text{PNCR}(u_1, u_2) = \exp\left(-\frac{|\mathbf{V}| - |I_1 \cap I_2|}{|\mathbf{V}|}\right) \quad (1.12)$$

Note, $|\mathbf{V}|$ is the number of all items.

Amer defined JacEDS as follows:

$$\text{JacEDS}(u_1, u_2) = \frac{|I_1 \cap I_2| + |\bar{I}_1 \cap \bar{I}_2|}{|\mathbf{V}|} \quad (1.13)$$

Where $\bar{I}_1$ and $\bar{I}_2$ are the sets of items that user 1 and user 2 do not rate, respectively. The reasonable proof of JacEDS is that JacEDS proves traditional Jaccard by concerning both items rated by users and items not rated by users whereas Jaccard only focuses on rated items. For example, given two rating vectors $u_3 = (r_{31}=4, r_{32}=1, r_{33}=5, r_{34}=5)$, and $u_4 = (r_{41}=1, r_{42}=2, r_{43}=?, r_{44}=?)$ from table 0.1, we have $I_3 = \{1, 2, 3, 4\}$, $I_4 = \{1, 2\}$, $\bar{I}_3 = \{\}$, $\bar{I}_4 = \{3, 4\}$, and $|\mathbf{V}| = 4$. Therefore, JacEDS of u_3 and u_4 is:

$$\text{JacEDS}(u_3, u_4) = \frac{|I_3 \cap I_4| + |\bar{I}_3 \cap \bar{I}_4|}{|\mathbf{V}|} = \frac{2 + 0}{4} = 0.5$$

Where $\bar{I}_1$ and $\bar{I}_2$ are the sets of items that user 1 and user 2 do not rate, respectively. Nguyen proposed dual Jaccard that balances the commonality among rated items and the commonality among non-rated items. Let P_1 and P_2 be the sets of items on which user 1 and user 2 gave relevant (positive) ratings, dual Jaccard is defined as follows:

$$\text{JacDual}(u_1, u_2) = k * \left(\frac{|P_1 \cap P_2|}{|P_1 \cup P_2|} + \frac{N - |P_1 \cup P_2|}{N - |P_1 \cap P_2|} \right) \quad (1.14)$$

Where k is called concerning coefficient or reinforcement factor which is now defined as a half of Jaccard measure.

$$k = 0.5 * \text{Jaccard}(u_1, u_2) = 0.5 \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|}$$

Note, N is the total number of items that both users rated. In equation of JacDual measure, the expression $\frac{|P_1 \cap P_2|}{|P_1 \cup P_2|}$ is exactly traditional Jaccard and the expression $\frac{N - |P_1 \cup P_2|}{N - |P_1 \cap P_2|}$ implies the similarity in rating between two users with regards to implicit inversion. Indeed, the term $N - |P_1 \cup P_2|$ indicates the number of common rated items in implicit inversion whereas the term $N - |P_1 \cap P_2|$ indicates cardinality of the union of rated items in implicit inversion. Therefore,

the concept “duality” in JacDual measure implies the duality of explicit Jaccard and implicit Jaccard. For example, given two rating vectors $u_3 = (r_{31}=4, r_{32}=1, r_{33}=5, r_{34}=5)$, and $u_4 = (r_{41}=1, r_{42}=2, r_{43}=?, r_{44}=?)$ from table 0.1, we have $I_3 = \{1, 2, 3, 4\}$, $I_4 = \{1, 2\}$, $\bar{I}_3 = \{\}$, $\bar{I}_4 = \{3, 4\}$, $P_3 = \{1, 3\}$, $P_4 = \{\}$, and $N = 4$. Therefore, dual Jaccard of u_3 and u_4 is:

$$\text{JacDual}(u_3, u_4) = 0.5 \frac{|I_3 \cap I_4|}{|I_3 \cup I_4|} \left(\frac{|P_3 \cap P_4|}{|P_3 \cup P_4|} + \frac{N - |P_3 \cup P_4|}{N - |P_3 \cap P_4|} \right) = 0.5 * \frac{2}{4} * \left(\frac{0}{2} + \frac{4-2}{4-0} \right) = 0.125$$

Al-Shamri (Al-Shamri, 2021) applied hyperbolic tangent function into improving Jaccard measure so as to propose a measure called JacHT. However, before mentioning JacHT, it is necessary to describe a so-called active-driven Jaccard where Jaccard is focused on active user in recommendation process who receives recommended list of items. Particularly, suppose user 1 named u_1 is active user, the active-driven Jaccard called Jacative is defined as follows (Al-Shamri, 2021, p. 4):

$$\text{Jactive}(u_1, u_2) = \frac{|I_1 \cap I_2|}{|I_1|}$$

Al-Shamri applied hyperbolic tangent function into Jacative to define **JacHT** as follows:

$$\begin{aligned} \text{JacHT}(u_1, u_2) &= h(\text{Jactive}(u_1, u_2)) \\ &= \frac{\exp(\text{Jactive}(u_1, u_2)) - \exp(-\text{Jactive}(u_1, u_2))}{\exp(\text{Jactive}(u_1, u_2)) + \exp(-\text{Jactive}(u_1, u_2))} \end{aligned} \quad (1.15)$$

Where $h(\cdot)$ is hyperbolic tangent function:

$$h(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

2. Cosine

Cosine which is the most popular similarity measure is specified as follows:

$$\text{sim}(u_1, u_2) = \cos(u_1, u_2) = \frac{u_1 \cdot u_2}{|u_1||u_2|} = \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j})^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j})^2}} \quad (2.1)$$

Where $|u_1|$ and $|u_2|$ are lengths of u_1 and u_2 , respectively whereas $u_1 \cdot u_2$ is dot product (scalar product) of u_1 and u_2 , respectively. If all ratings are non-negative, range of cosine measure is from 0 to 1. If it is equal to 0, two users are totally different. If it is equal to 1, two users are identical. By following the ideology of Jaccard measure, cosine measure is modified as follows:

$$\text{COJ}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{\sqrt{\sum_{j \in I_1} (r_{1j})^2} \sqrt{\sum_{j \in I_2} (r_{2j})^2}} \quad (2.2)$$

Traditional cosine is only determined by the common set of items which are rated by both users but COJ is more general by concerning items which are rated by at least one user. Amer proposed a so-called numerical nearby similarity measure (NNSM) which is also more general than cosine but it saves computation cost better than COJ. NNSM developed by Amer and implemented by Nguyen is defined as follows:

$$\text{NNSMCo}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{|I_1| \sum_{j \in I_1} r_{1j} + |I_2| \sum_{j \in I_2} r_{2j}} \quad (2.3)$$

Let r_m be median of rating values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$. The normalized cosine measure (CON) (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 158) is defined as follows:

$$\text{CON}(u_1, u_2) = \text{CPC}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} (r_{1j} - r_m)(r_{2j} - r_m)}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - r_m)^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - r_m)^2}}$$

Obviously, CON measure is constrained Pearson correlation (CPC) mentioned later.

Let $v_j = (r_{1j}, r_{2j}, \dots, r_{mj})$ be vector of rating values that item j receives from m users, for example. The mean of v_j is:

$$\bar{v}_j = \frac{1}{m} \sum_{i=1}^m r_{ij}$$

Adjusted cosine measure (COD) is defined as follows:

$$\text{COD}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{v}_j)(r_{2j} - \bar{v}_j)}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{v}_j)^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - \bar{v}_j)^2}} \quad (2.4)$$

Jaccard can be combined with any measure. For instance, CosineJ is combinations of Jaccard and cosine as follows:

$$\begin{aligned} \text{CosineJ}(u_1, u_2) &= \text{cosine}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \\ &= \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{\sqrt{\sum_{j \in I_1} (r_{1j})^2} \sqrt{\sum_{j \in I_2} (r_{2j})^2}} * \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} \end{aligned} \quad (2.5)$$

3. Pearson

Pearson correlation is another popular similarity measure besides cosine, which is defined as follows (Sarwar, Karypis, Konstan, & Riedl, 2001, p. 290):

$$\text{Pearson}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)(r_{2j} - \bar{u}_2)}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - \bar{u}_2)^2}} \quad (3.1)$$

Where $\bar{u}_1$ and $\bar{u}_2$ are mean values of u_1 and u_2 , respectively.

$$\bar{u}_1 = \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \bar{u}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j}$$

The range of Pearson measure is from -1 to 1 . If it is equal to -1 , two users are totally opposite. If it is equal to 1 , two users are identical. Pearson measure is sample correlation coefficient in statistics. Pearson measure has some variants. Constrained Pearson correlation (CPC) measure considers impact of positive and negative ratings by using median r_m instead of using the means; for example, if rating values range from 1 to 5 , the median is $r_m = (1+5) / 2 = 3$. CPC measure is defined as follows (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 158):

$$\text{CPC}(u_1, u_2) = \text{CON}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} (r_{1j} - r_m)(r_{2j} - r_m)}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - r_m)^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - r_m)^2}} \quad (3.2)$$

The similarity will be significant if both users rated more common items. Weight Pearson correlation (WPC) measure and sigmoid Pearson correlation (SPC) measure concern how much common items are. WPC is defined as follows (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 158):

$$\text{WPC}(u_1, u_2) = \begin{cases} \text{Pearson}(u_1, u_2) * \frac{|I|}{H}, & \text{if } |I| \leq H \\ \text{Pearson}(u_1, u_2), & \text{otherwise} \end{cases} \quad (3.3)$$

SPC is defined as follows (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 158):

$$\text{SPC}(u_1, u_2) = \text{Pearson}(u_1, u_2) * \frac{1}{1 + \exp\left(-\frac{|I|}{2}\right)} \quad (3.4)$$

Where H is a threshold and it is often set to be 50 (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 158).

In the determinant of traditional Pearson measures, both vectors u_1 and u_2 are taken account of common items $I_1 \cap I_2$. Alternately, Ayub et al. (Ayub M. , et al., 2019) calculated module of each vector on its full rated item regardless of the other's rated items. Their measure is called improved Pearson correlation (IPC) is defined as follows (Ayub M. , et al., 2019, p. 7):

$$\text{IPC}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} (r_{1j} - r_m)(r_{2j} - r_m)}{\sqrt{\sum_{j \in I_1} (r_{1j} - r_m)^2} \sqrt{\sum_{j \in I_2} (r_{2j} - r_m)^2}} \quad (3.5)$$

Ayub et al. (Ayub M. , et al., 2019) also proposed a new improvement of Pearson measure by combining IPC and rating preference behavior (RPB) measure. Thus, their proposed measure is called improved Pearson correlation with rating preference behavior (IPWR), which is defined as follows (Ayub M. , et al., 2019, p. 8):

$$\text{IPWR}(u_1, u_2) = \alpha \text{RPB}(u_1, u_2) + \beta \text{IPC}(u_1, u_2) \quad (3.6)$$

Where α and β are two weights which are applied to RPB and IPC. RPB measure is trigonometric cosine function of statistical deviations, specified as follows (Ayub M. , et al., 2019, p. 6):

$$\text{RPB}(u_1, u_2) = \cos(|\bar{u}_1 - \bar{u}_2| |\text{var}(u_1) - \text{var}(u_2)|)$$

Note, $\bar{u}_i$ and $\text{var}(u_i)$ are mean and variance of u_i , respectively.

$$\bar{u}_i = \frac{1}{|I_i|} \sum_{j \in I_i} r_{ij}, \text{var}(u_i) = \frac{1}{|I_i|} \sum_{j \in I_i} (r_{ij} - \bar{u}_i)^2$$

In practice, the variance of u_i can be replaced by the standard deviation of u_i as follows (Ayub M. , et al., 2019, p. 6):

$$\text{RPB}(u_1, u_2) = \cos(|\bar{u}_1 - \bar{u}_2| |\text{sd}(u_1) - \text{sd}(u_2)|)$$

$$\text{sd}(u_i) = \sqrt{\text{var}(u_i)}$$

PearsonJ is combinations of Jaccard and Pearson as follows:

$$\begin{aligned} \text{PearsonJ}(u_1, u_2) &= \text{Pearson}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \\ &= \frac{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)(r_{2j} - \bar{u}_2)}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - \bar{u}_2)^2}} * \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} \end{aligned} \quad (3.7)$$

Choi and Suh (Choi & Suh, 2013) proposed a so-called PC measure which is Pearson measure weighted by similarities of items. In other words, PC measure combines similarities of users and items (Patra, Launonen, Ollikainen, & Nandi, 2015, p. 4). The ideology is excellent. PC measure can be applied into any foundation measures. Each factor in PC measure is weighted by a similarity of active item and another item. Suppose it is necessary to estimate rating values of active item k , PC measure (Choi & Suh, 2013, p. 148) is defined as follows:

$$\text{PC}_k(u_1, u_2) = \frac{\sum_{j \in I} \left(\left(\text{sim}(v_k, v_j) \right)^2 (r_{1j} - \bar{u}_1)(r_{2j} - \bar{u}_2) \right)}{\sqrt{\sum_{j \in I_1} \left(\text{sim}(v_k, v_j)(r_{1j} - \bar{u}_1) \right)^2} \sqrt{\sum_{j \in I_2} \left(\text{sim}(v_k, v_j)(r_{2j} - \bar{u}_2) \right)^2}} \quad (3.8)$$

Where $\text{sim}(v_k, v_j)$ is similarity of the active item k and item j . Note, $\text{sim}(v_k, v_j)$ can be calculated by any measures here. The $\bar{u}_1$ and $\bar{u}_2$ are mean values of u_1 and u_2 , respectively.

$$\bar{u}_1 = \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \bar{u}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j}$$

Experimental results proved that PC is an effective measure.

4. MSD

Mean squared difference (MSD) is defined as inverse of distance between two vectors. Let MAX be maximum value of ratings, MSD is calculated as follows (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 158):

$$\text{MSD}(u_1, u_2) = 1 - \frac{\sum_{j \in I} \left(\frac{r_{1j} - r_{2j}}{\text{MAX}} \right)^2}{|I|} \quad (4.1)$$

Another variant of MSD is specified by some authors as follows:

$$\text{MSD}(u_1, u_2) = \frac{1}{1 + \frac{1}{|I|} \sum_{j \in I} (r_{1j} - r_{2j})^2} \quad (4.2)$$

MSD measure combines with Jaccard measure, which derives MSDJ measure as follows (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 158):

$$\text{MSDJ}(u_1, u_2) = \text{MSD}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \quad (4.3)$$

5. SRC

When rating values are converted into ranks, Spearman's Rank Correlation (SRC) is defined as follows (Hyung, 2008, p. 39):

$$\text{SRC}(u_1, u_2) = 1 - \frac{6 \sum_{j \in I} d_j^2}{|I|(|I|^2 - 1)} \quad (5.1)$$

Where d_j is difference between two ranks on item j given by user 1 and user 2.

$$d_j = \text{rank}_{1j} - \text{rank}_{2j}$$

Note, it is easy to convert ratings values to ranks. For example, suppose rating values (bins) are 5, 6, 7, 8, 9 then, we have rank 1 (for value 9), rank 2 (for value 8), rank 3 (for value 7), rank 4 (for value 6), and rank 5 (for value 5). If user 1 rates value 9 to item j , we have $\text{rank}_{1j} = 1$. The larger the value is, the smaller (higher) the rank is.

6. PIP

Ahn (Hyung, 2008) proposed a heuristic measure to solve cold-starting problem which relates to missing data in which there is not enough information to calculate similarities between rating vectors (Hyung, 2008, p. 39). The measure called PIP measure based on concept of "agreement" in rating. If both user 1 and user 2 like or dislike the same item, it is called that they have a rating "agreement" on such item. Let r_{1j} and r_{2j} be ratings of user 1 and user 2 on item j , respectively, the agreement (Hyung, 2008, p. 43) of them is defined as follows:

$$\text{agree}(r_{1j}, r_{2j}) = \begin{cases} \text{true if } (r_{1j} > r_m \text{ and } r_{2j} > r_m) \\ \text{true if } (r_{1j} < r_m \text{ and } r_{2j} < r_m) \\ \text{false otherwise} \end{cases}$$

Note, r_m be median of rating values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$. PIP measure (Hyung, 2008, p. 42) is sum of products of triples Proximity, Impact, and Popularity.

$$\text{PIP}(u_1, u_2) = \sum_{j \in I_1 \cap I_2} \text{Proximity}(r_{1j}, r_{2j}) * \text{Impact}(r_{1j}, r_{2j}) * \text{Popularity}(r_{1j}, r_{2j}) \quad (6.1)$$

Proximity (Hyung, 2008, p. 43) indicates similarity of two ratings, based on agreement and distance between them. The distance is increased twice as a penalty if such two ratings are not agreed.

$$\text{Proximity}(r_{1j}, r_{2j}) = \begin{cases} \left((2(r_{\max} - r_{\min}) + 1) - |r_{1j} - r_{2j}| \right)^2 & \text{if agree}(r_{1j}, r_{2j}) = \text{true} \\ \left((2(r_{\max} - r_{\min}) + 1) - 2|r_{1j} - r_{2j}| \right)^2 & \text{if agree}(r_{1j}, r_{2j}) = \text{false} \end{cases} \quad (6.2)$$

Where $r_{\min}$ and $r_{\max}$ are minimum rating value and maximum rating value, respectively. If two ratings are agreed, their impact (Hyung, 2008, p. 43) is proportional to difference between them and rating median. If two ratings are disagreed, their impact is inverse of such difference.

$$\text{Impact}(r_{1j}, r_{2j}) = \begin{cases} (|r_{1j} - r_m| + 1) * (|r_{2j} - r_m| + 1) & \text{if agree}(r_{1j}, r_{2j}) = \text{true} \\ 1 & \\ \frac{1}{(|r_{1j} - r_m| + 1) * (|r_{2j} - r_m| + 1)} & \text{if agree}(r_{1j}, r_{2j}) = \text{false} \end{cases} \quad (6.3)$$

Popularity (Hyung, 2008, p. 43) indicates difference between ratings given by active users and the average rating.

$$\text{Popularity}(r_{1j}, r_{2j}) = \begin{cases} 1 + \left(\frac{r_{1j} + r_{2j}}{2} - \mu_j \right)^2 & \text{if } (r_{1j} > \mu_j \text{ and } r_{2j} > \mu_j) \\ 1 + \left(\frac{r_{1j} + r_{2j}}{2} - \mu_j \right)^2 & \text{if } (r_{1j} < \mu_j \text{ and } r_{2j} < \mu_j) \\ 1 & \text{otherwise} \end{cases} \quad (6.4)$$

Note, μ_j is average rating of item j , which is same mean of rating values of item j . Experimental results proved that cold-starting problem is solved well by PIP measure (Hyung, 2008, p. 47).

Manochandar and Punniyamoorthy (Manochandar & Punniyamoorthy, 2020) proposed the modified PIP (MPIP) measure which is still the sum of products of triples Proximity, Impact, and Popularity but they improved such triples. For instance, the Proximity quantity is improved as follows (Manochandar & Punniyamoorthy, 2020, p. 595):

$$\text{Proximity2}(r_{1j}, r_{2j}) = \begin{cases} \left(\frac{|r_{1j} - r_{2j}| - \frac{r_m^+ + r_m^-}{2}}{r_{\max} - r_{\min}} \right)^2 & \text{if agree}(r_{1j}, r_{2j}) = \text{true} \\ \delta \left(\frac{1}{\frac{|r_{1j} - r_{2j}|}{r_{\max} - r_{\min}}} \right)^2 & \text{if agree}(r_{1j}, r_{2j}) = \text{false} \end{cases} \quad (6.5)$$

Note, r_m is the median; for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$. By similar inference, r_m^+ is the specified median that is larger than r_m ; for example, if rating values range from 1 to 5 then, $r_m^+ = (4+5) / 2 = 4.5$. Thus, r_m^- is the specified median

that is smaller than r_m ; for example, if rating values range from 1 to 5 then, $r_m^- = (1+2) / 2 = 1.5$. The parameter δ is rating scale as follows (Manochandar & Punniyamoorthy, 2020, p. 595):

$$\delta = \begin{cases} 0.75 & \text{if } |r_{1j} - r_{2j}| > r_m \\ 0.5 & \text{if } |r_{1j} - r_{2j}| = r_m \\ 0.25 & \text{otherwise} \end{cases}$$

Manochandar and Punniyamoorthy modified the Impact quantity as follows (Manochandar & Punniyamoorthy, 2020, p. 596):

$$\text{Impact2}(r_{1j}, r_{2j}) = \begin{cases} \exp\left(-\frac{1}{(|r_{1j} - r_m| + 1) * (|r_{2j} - r_m| + 1)}\right) & \text{if } \text{agree}(r_{1j}, r_{2j}) = \text{true} \\ \frac{1}{(|r_{1j} - r_m| + 1) * (|r_{2j} - r_m| + 1)} & \text{if } \text{agree}(r_{1j}, r_{2j}) = \text{false} \end{cases} \quad (6.6)$$

They also modified the Popularity quantity as follows (Manochandar & Punniyamoorthy, 2020, p. 596):

$$\text{Popularity2}(r_{1j}, r_{2j}) = \begin{cases} \log_{10}\left(2 + \left(\frac{r_{1j} + r_{2j}}{2} - \mu_j\right)^2\right) & \text{if } (r_{1j} > \mu_j \text{ and } r_{2j} > \mu_j) \\ & \text{or } (r_{1j} < \mu_j \text{ and } r_{2j} < \mu_j) \\ 0.3010 & \text{otherwise} \end{cases} \quad (6.7)$$

Note, μ_j is average rating of item j . As a result, MPIP is specified as follows:

$$\text{MPIP}(u_1, u_2) = \sum_{j \in I_1 \cap I_2} \text{Proximity2}(r_{1j}, r_{2j}) * \text{Impact2}(r_{1j}, r_{2j}) * \text{Popularity2}(r_{1j}, r_{2j}) \quad (6.8)$$

7. PSS and NHMS

Liu et al. (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 156) proposed a new similarity measure called NHMS to improve recommendation task in which only few ratings are available. Their NHMS measure (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 160) is based on sigmoid function and the improved PIP measure as PSS (*Proximity – Significance – Singularity*). PSS similarity is calculated as follows (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 160):

$$\text{PSS}(u_1, u_2) = \sum_{j \in I} \text{Proximity}(r_{1j}, r_{2j}) * \text{Significance}(r_{1j}, r_{2j}) * \text{Singularity}(r_{1j}, r_{2j}) \quad (7.1)$$

Where, $I = I_1 \cap I_2$ is intersection set of I_1 and I_2 . The proximity factor determines similarity of two ratings, based on distance between them; such distance is as less as better. The significance factor determines similarity of two ratings, based on distance from them to rating median; such distance is as more as better. The singularity factor determines similarity of two ratings, based on difference between them and other ratings; such difference is as less as better. Followings are equations of these factors based on sigmoid function (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 161).

$$\begin{aligned}\text{Proximity}(r_{1j}, r_{2j}) &= 1 - \frac{1}{1 + \exp(-|r_{1j} - r_{2j}|)} \\ \text{Significance}(r_{1j}, r_{2j}) &= \frac{1}{1 + \exp(-|r_{1j} - r_m||r_{2j} - r_m|)} \\ \text{Singularity}(r_{1j}, r_{2j}) &= 1 - \frac{1}{1 + \exp\left(-\left|\frac{r_{1j} + r_{2j}}{2} - \mu_j\right|\right)}\end{aligned}$$

Note, r_m be median of rating values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$ whereas μ_j is rating mean of item j . Liu et al. (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 161) also considered the similarity between two users via URP measure as follows:

$$\text{URP}(u_1, u_2) = 1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2||\sigma_1 - \sigma_2|)} \quad (7.2)$$

Where μ_1 and μ_2 are rating means of user 1 and user 2, respectively and σ_1 and σ_2 are rating standard deviations of user 1 and user 2, respectively.

$$\begin{aligned}\mu_1 &= \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \mu_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j} \\ \sigma_1 &= \sqrt{\frac{1}{|I_1|} \sum_{j \in I_1} (r_{1j} - \mu_1)^2}, \sigma_2 = \sqrt{\frac{1}{|I_2|} \sum_{j \in I_2} (r_{2j} - \mu_2)^2}\end{aligned}$$

PSS associated with Jaccard produces a so-called PSSJ measure as follows:

$$\text{PSSJ}(u_1, u_2) = \text{PSS}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \quad (7.3)$$

Where Jaccard is specified by equation 1.1, respectively. Liu et al. (Liu, Hu, Mian, Tian, & Zhu, 2013, p. 161) proposed a new heuristic similarity model (NHSM) as triple product of PSS measure, URP measure, and Jaccard2 measure.

$$\text{NHSM}(u_1, u_2) = \text{PSS}(u_1, u_2) * \text{URP}(u_1, u_2) * \text{Jaccard2}(u_1, u_2) \quad (7.4)$$

In general, Liu et al. (Liu, Hu, Mian, Tian, & Zhu, 2013) aim to alleviate the problem of few rated common items via their NHSM measure. From experimental result, NHSM gave out excellent estimation.

8. BCF

Patra et al. (Patra, Launonen, Ollikainen, & Nandi, 2015, p. 143) proposed a new similarity measure called BCF for CF, which uses all ratings made by a pair of users. Proposed measure finds importance of each pair of rated items by exploiting Bhattacharyya (BC) similarity. The BC similarity, which is core of their own measure, measures the similarity between two distributions. So, these distributions are estimated as the number of uses rated on given item. In general, Patra et al. (Patra, Launonen, Ollikainen, & Nandi, 2015, p. 5) combined BC similarity and the local similarity where the local similarity relates to Pearson correlation. It is necessary to survey BC similarity. Bin is a terminology indicating domain of rating values, for example, if rating values range from 1 to 5, we have bins: 1, 2, 3, 4, 5. Let m be the number of bins, given items i and j , item BC coefficient for items is calculated as follows (Patra, Launonen, Ollikainen, & Nandi, 2015, p. 5):

$$\text{bc}(i, j) = \sum_{h=1}^m \sqrt{\frac{\#h_i \#h_j}{\#i \#j}} \quad (8.1)$$

Note, $\#i$ and $\#j$ are the numbers of users who rated items i and j , respectively whereas $\#h_i$ and $\#h_j$ are numbers of users who gave rating value h on items i and j , respectively. So, item BC coefficient concerns two items. In table 0.2, rating vectors of item 3 and item 4 are $v_3 = (1, 2,$

5, ?) and $v_4 = (5, 4, 5, ?)$, respectively with note that rating values range from 1 to 5 and so we have:

$$bc(v_3, v_4) = \sqrt{\frac{1}{4} \frac{0}{4}} + \sqrt{\frac{1}{4} \frac{0}{4}} + \sqrt{\frac{0}{4} \frac{0}{4}} + \sqrt{\frac{0}{4} \frac{1}{4}} + \sqrt{\frac{1}{4} \frac{2}{4}} \cong 0.35$$

According to Patra et al. (Patra, Launonen, Ollikainen, & Nandi, 2015, p. 5), user BC similarity is sum of products of item BC coefficients and local similarities as follows:

$$BC(u_1, u_2) = \sum_{i \in I_1} \sum_{j \in I_2} bc(i, j) loc(r_{1i}, r_{2j}) \quad (8.2)$$

The local similarity is calculated as a part of constrained Pearson coefficient (CPC) as follows:

$$loc(r_{1i}, r_{2j}) = \frac{(r_{1i} - r_m)(r_{2j} - r_m)}{\sqrt{\sum_{k \in I_1} (r_{1k} - r_m)^2} \sqrt{\sum_{k \in I_2} (r_{2k} - r_m)^2}}$$

Note, r_m be median of rating values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$. Patra et al. (Patra, Launonen, Ollikainen, & Nandi, 2015, p. 5) proposed Bhattacharyya similarity in CF (BCF) as sum of user BC similarity and Jaccard measure as follows:

$$BCFJ(u_1, u_2) = Jaccard(u_1, u_2) + BC(u_1, u_2) \quad (8.3)$$

Singh et al. (Singh, Sinha, & Choudhury, 2022) improved BCF measure by applying BCF measure into both user vector and item vector. Let $BC(v_1, v_2)$ be BC similarity of item v_1 and item v_2 as follows:

$$BC(v_1, v_2) = \sum_{k \in U_1} \sum_{l \in U_2} bc(k, l) loc(r_{k1}, r_{l2})$$

Note, $bc(k, l)$ is user BC coefficient for users:

$$bc(k, l) = \sum_{h=1}^m \sqrt{\frac{\#h_k}{\#k} \frac{\#h_l}{\#l}}$$

Where $\#k$ and $\#l$ are the numbers of items which are rated by users k and l , respectively whereas $\#h_k$ and $\#h_l$ are numbers of items which are rated with rating value h by users k and l , respectively. The local similarity $loc(r_{k1}, r_{l2})$ with regard to items is specified as follows:

$$loc(r_{k1}, r_{l2}) = \frac{(r_{k1} - r_m)(r_{l2} - r_m)}{\sqrt{\sum_{p \in U_1} (r_{kp} - r_m)^2} \sqrt{\sum_{p \in U_2} (r_{lp} - r_m)^2}}$$

Singh et al. (Singh, Sinha, & Choudhury, 2022) proposed **BCF2** measure by browsing BC similarity over both users and items, as follows:

$$BCF2(u_1, u_2) = \frac{\sum_{i \in I_1} \sum_{j \in I_2} BC(v_i, v_j) loc(r_{1i}, r_{2j})}{\sum_{i \in I_1} \sum_{j \in I_2} BC(v_i, v_j)} \quad (8.4)$$

Where local similarity is improved by mutual associating with BC similarity as follows:

$$loc(r_{1i}, r_{2j}) = BC(u_1, u_2) \frac{(r_{1i} - r_m)(r_{2j} - r_m)}{\sqrt{\sum_{k \in I_1} (r_{1k} - r_m)^2} \sqrt{\sum_{k \in I_2} (r_{2k} - r_m)^2}} \quad (8.5)$$

Obviously, Singh et al. (Singh, Sinha, & Choudhury, 2022) improved BCF measure by operating BCF measure by itself. Singh et al. (Singh, Sinha, & Choudhury, 2022) also added Jaccard measure into their measure BCF2 for improving BCF2 more, as BCF2J measure as follows:

$$BCF2J(u_1, u_2) = Jaccard(u_1, u_2) + BCF2(u_1, u_2) \quad (8.6)$$

9. MMD

Suryakant and Mahara (Suryakant & Mahara, 2016) proposed a so-called Cosine-Jaccard-Mean Measure of Divergence (CjacMD) based on Mean Measure of Divergence (MMD) to solve the problem of sparse rating matrix. Because MMD measure takes advantages of statistical aspects, it can alleviate sparsity. MMD focuses on personal habits which are ignored by nonstatistical measures (Suryakant & Mahara, 2016, p. 453). Recall that bin is a terminology indicating domain of rating values, for example, if rating values range from 1 to 5, we have bins: 1, 2, 3, 4, 5. Let $X = (x_1, x_2, \dots, x_b)$ and $Y = (y_1, y_2, \dots, y_b)$ be count vectors of user 1 and user 2, respectively where x_j (y_j) is the number of items to which user 1 (user 2) gives bin j with note that b is the number of bins. For example, rating vectors of user 1 and user 2 in table 0.1 are $u_1 = (1, 2, 1, 5)$ and $u_2 = (2, 1, 2, 4)$, respectively with note that rating values ranges from 1 to 5. We have:

$$\begin{aligned} X &= (x_1 = 2, x_2 = 1, x_3 = 0, x_4 = 0, x_5 = 1) \\ Y &= (y_1 = 1, y_2 = 2, y_3 = 0, y_4 = 1, y_5 = 0) \end{aligned}$$

MMD measure is defined as follows (Suryakant & Mahara, 2016, p. 453), (Harris & Sjøvold, 2018, p. 87):

$$\text{MMD}(u_1, u_2) = \frac{1}{1 + \frac{1}{b} \sum_{j=1}^b \left((\theta_{1j} - \theta_{2j})^2 - \frac{1}{0.5 + x_j} - \frac{1}{0.5 + y_j} \right)} \quad (9.1)$$

Where θ_{1*} and θ_{2*} are Grewal's transformations (Harris & Sjøvold, 2018, p. 85) of X and Y , respectively.

$$\begin{aligned} \theta_{1j} &= \frac{1}{\sin \left(1 - \frac{2x_j}{|I_1|} \right)} \\ \theta_{2j} &= \frac{1}{\sin \left(1 - \frac{2y_j}{|I_2|} \right)} \end{aligned}$$

In fact, CjacMD (Suryakant & Mahara, 2016, p. 453) combines three other measures such as cosine, Jaccard, and MMD together.

$$\text{CjacMD} = \cos(u_1, u_2) + \text{Jaccard}(u_1, u_2) + \text{MMD}(u_1, u_2) \quad (9.2)$$

Experimental result proved that CjacMD model is effective similarity model.

10. Triangle

Sun et al. (Sun, et al., 2017) proposed a so-called Triangle similarity measure which considers both angle and lengths of rating vectors. For instance, given two user vectors u_1 and u_2 are considered as two vector $OA = u_1$ and $OB = u_2$ and hence, OAB forms a triangle. TS measure is ratio of the length $|AB|$ to the sum of lengths $|OA| + |OB|$. Of course, $|AB|$ is always less than or equal to $|OA| + |OB|$ according to triangle inequality. The idea is excellent. TS measure (Sun, et al., 2017, p. 6) is defined as follows:

$$\begin{aligned} \text{Triangle}(u_1, u_2) &= 1 - \frac{|AB|}{|OA| + |OB|} = 1 - \frac{|u_1 - u_2|}{|u_1| + |u_2|} \\ &= 1 - \frac{\sqrt{\sum_{j \in I} (r_{1j} - r_{2j})^2}}{\sqrt{\sum_{j \in I} r_{1j}^2} + \sqrt{\sum_{j \in I} r_{2j}^2}} \end{aligned} \quad (10.1)$$

Sun et al. also combined Triangle measure and Jaccard measure to form a new measure called Triangle multiplying Jaccard (TMJ) measure. The integrated TMJ (Sun, et al., 2017, p. 6) is defined as follows:

$$\text{TMJ}(u_1, u_2) = \text{Triangle}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \quad (10.2)$$

Experimental result proved that TMJ is effective measure.

11. Feng

To solve the problem of sparse rating matrix, Feng et al. (Feng, Fengs, Zhang, & Peng, 2018) proposed a new model of similarity which includes three parts such as S_1 , S_2 , and S_3 . The S_1 (Feng, Fengs, Zhang, & Peng, 2018, p. 6) is normal similarity and they choose cosine as S_1 .

$$S_1(u_1, u_2) = \begin{cases} \text{cosine}(u_1, u_2) & \text{if sparsity} < \rho \\ \text{COJ}(u_1, u_2) & \text{otherwise} \end{cases}$$

Where ρ is sparsity threshold which is proposed by Feng et al. The S_2 (Feng, Fengs, Zhang, & Peng, 2018, p. 6) punishes user pairs whose co-rated items are few.

$$S_2(u_1, u_2) = \frac{1}{1 + \exp\left(-\frac{|I_1 \cap I_2|^2}{|I_1||I_2|}\right)}$$

The S_3 (Feng, Fengs, Zhang, & Peng, 2018, p. 6) focuses on statistical feature of user ratings, which reflects essential user favorites. S_3 is aforementioned URP measure.

$$S_3(u_1, u_2) = \text{URP}(u_1, u_2) = 1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2| |\sigma_1 - \sigma_2|)}$$

Where μ_1 and μ_2 are rating means of user 1 and user 2, respectively and σ_1 and σ_2 are rating standard deviations of user 1 and user 2, respectively. The similarity model of Feng et al. (Feng, Fengs, Zhang, & Peng, 2018, p. 5) is product of S_1 , S_2 , and S_3 as follows:

$$\text{Feng}(u_1, u_2) = S_1(u_1, u_2) * S_2(u_1, u_2) * S_3(u_1, u_2) \quad (11.1)$$

Experimental result proved that Feng model is effective similarity model.

12. Mu

Mu et al. (Mu, Xiao, Tang, Luo, & Yin, 2019) combined local measures (Pearson and Jaccard) with global measure to solve the problem of sparse rating matrix. The global measure is Hellinger (Hg) distance which estimates similarity of two probabilistic distributions. In fact, Hg is inverse of BC coefficient in discrete distributions as follows (Mu, Xiao, Tang, Luo, & Yin, 2019, p. 419):

$$\text{Hg}(u_1, u_2) = 1 - \text{bc}(u_1, u_2) = 1 - \sum_{h=1}^m \sqrt{\frac{\#h_1}{\#1} \frac{\#h_2}{\#2}} \quad (12.1)$$

Note, #1 and #2 are the numbers of item which are rated by user 1 and user 2, respectively whereas $\#h_1$ and $\#h_2$ are numbers of items which receive rating value h from user 1 and user 2, respectively. For example, rating vectors of user 1 and user 2 in table 0.1 are $u_1 = (1, 2, 1, 5)$ and $u_2 = (2, 1, 2, 4)$, respectively with note that rating values range from 1 to 5 and so we have:

$$\text{Hg}(u_1, u_2) = 1 - \left(\sqrt{\frac{2}{4} \frac{1}{4}} + \sqrt{\frac{1}{4} \frac{2}{4}} + \sqrt{\frac{0}{4} \frac{0}{4}} + \sqrt{\frac{0}{4} \frac{1}{4}} + \sqrt{\frac{1}{4} \frac{0}{4}} \right) \cong 0.29$$

Given weight α , the Mu measure (Mu, Xiao, Tang, Luo, & Yin, 2019, p. 419) combines Pearson, Jaccard, and Hg as follows:

$$\text{Mu}(u_1, u_2) = \alpha * \text{Pearson}(u_1, u_2) + (1 - \alpha) * (\text{Hg}(u_1, u_2) + \text{Jaccard}(u_1, u_2)) \quad (12.2)$$

Experimental result proved that Mu measure is effective similarity model.

13. SMTP

Similarity Measure for Text Processing (SMTP) was developed by Lin, Jiang, and Lee (Lin, Jiang, & Lee, 2013), originally used for computing the similarity between two documents in text processing. Here documents are considered as rating vectors. Given two rating vectors u_1

$= (r_{11}, r_{12}, \dots, r_{1n})$ and $u_2 = (r_{21}, r_{22}, \dots, r_{2n})$, the function F of u_1 and u_2 is defined as follows (Lin, Jiang, & Lee, 2013, p. 1577):

$$F(u_1, u_2) = \frac{\sum_{j=1}^n A(r_{1j}, r_{2j})}{\sum_{j=1}^n B(r_{1j}, r_{2j})} \quad (13.1)$$

Where (Lin, Jiang, & Lee, 2013, p. 1577),

$$A(r_{1j}, r_{2j}) = \begin{cases} 0.5 \left(1 + \exp \left(- \left(\frac{r_{1j} - r_{2j}}{\sigma_j} \right)^2 \right) \right) & \text{if both } r_{1j} \text{ and } r_{2j} \text{ non - missing} \\ 0 & \text{if both } r_{1j} \text{ and } r_{2j} \text{ missing} \\ -\lambda & \text{otherwise} \end{cases}$$

$$B(r_{1j}, r_{2j}) = \begin{cases} 0 & \text{if both } r_{1j} \text{ and } r_{2j} \text{ missing} \\ 1 & \text{if both } r_{1j} \text{ and } r_{2j} \text{ non - missing} \end{cases}$$

Note that λ is the pre-defined number and σ_j is the standard deviation of rating values belonging to field j (item j). In this research, λ is set to be 0.5. Lin, Jiang, and Lee (Lin, Jiang, & Lee, 2013, p. 1577) defined SMTP measure based on function F as follows:

$$\text{SMTP}(u_1, u_2) = \frac{F(u_1, u_2) + \lambda}{1 + \lambda} \quad (13.2)$$

14. SMD

Given two rating vectors $u_1 = (r_{11}, r_{12}, \dots, r_{1n})$ and $u_2 = (r_{21}, r_{22}, \dots, r_{2n})$ of user 1 and user 2, respectively, in which some r_{ij} can be missing (empty). In binary representation, r_{ij} is converted into 1 if it is non-missing (rated) and otherwise, r_{ij} is converted into 0 if it is missing (not rated). Let N_{12} be the number of common values “1” in both u_1 and u_2 . Let N be the total number of all items under consideration; in this case, $N = n$. Let N_1 and N_2 be the numbers of values “1” of u_1 and u_2 , respectively. Let F be the number of differences between u_1 and u_2 ; for example, the fact that $r_{11} = 0$ and $r_{21} = 1$ contributes one difference to F . Amer defined a so-called SMD measure in binary representation as follows:

$$\text{SMD} = \frac{\left(1 - \frac{F}{N}\right) + \left(\frac{2N_{12}}{N_1 + N_2}\right)}{2} \quad (14.1)$$

Now we interpret SMD measure in context of rating matrix. Let I_1 and I_2 be set of indices of items that user 1 and user 2 rated, respectively, we have:

$$\begin{aligned} N_1 &= |I_1| \\ N_2 &= |I_2| \\ N_{12} &= |I_1 \cap I_2| \\ N &= |I_1 \cup I_2| \\ F &= N - |I_1 \cap I_2| = |I_1 \cup I_2| - |I_1 \cap I_2| \end{aligned}$$

Therefore, we obtain:

$$\begin{aligned} \text{SMD} &= \frac{\left(1 - \frac{|I_1 \cup I_2| - |I_1 \cap I_2|}{|I_1 \cup I_2|}\right) + \left(\frac{2|I_1 \cap I_2|}{|I_1| + |I_2|}\right)}{2} = \frac{\frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} + \frac{2|I_1 \cap I_2|}{|I_1| + |I_2|}}{2} \\ &= \left(\frac{0.5}{|I_1 \cup I_2|} + \frac{1}{|I_1| + |I_2|}\right) |I_1 \cap I_2| \end{aligned}$$

Let,

$$\text{SMD} = \left(\frac{0.5}{|I_1 \cup I_2|} + \frac{1}{|I_1| + |I_2|}\right) A$$

Where,

$$A = |I_1 \cap I_1|$$

Amer also defined another so-called *HSMD measure* in numerical representation in which values r_{ij} are kept in numerical values as rating values, as follows:

$$\text{HSMD} = 1 - \frac{R_1 R_2 + 1}{G} \quad (14.2)$$

Where, R_1 (R_2) is the sum of non-missing values r_{1j} (r_{2j}) of u_1 (u_2) such that respective values r_{2j} (r_{1j}) are missing.

$$R_1 = \sum_{\substack{r_{1j} \text{ non-missing} \\ r_{2j} \text{ missing}}} r_{1j} = \sum_{j \in I_1 \setminus I_2} r_{1j}$$

$$R_2 = \sum_{\substack{r_{2j} \text{ non-missing} \\ r_{1j} \text{ missing}}} r_{2j} = \sum_{j \in I_2 \setminus I_1} r_{2j}$$

Note, notation “ $\setminus$ ” denote complement operator in set theory. G is product of two sums of non-missing values for both r_1 and r_2 .

$$G = \left(\sum_{r_{1j} \text{ non-missing}} r_{1j} \right) \left(\sum_{r_{2j} \text{ non-missing}} r_{2j} \right) = \left(\sum_{j \in I_1} r_{1j} \right) \left(\sum_{j \in I_2} r_{2j} \right)$$

In general, measures SMD and HSMD are defined firstly for weight vectors of documents in information retrieval, in which every element of a vector is a weight which is product of term frequency (TF) and inverse document frequency (IDF). Here they are applied into CF. For example, given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$. Binary representations of these two vectors are (1, 1, 1, 1, 0, 1) and (1, 0, 0, 1, 1, 1). According to SMD measure, we have $N_{12}=3$, $N=6$, $F=3$, $N_1=5$, and $N_2=4$. Hence, SMD measure is calculated according to equation 14.1, as follows:

$$\text{SMD} = \frac{\left(1 - \frac{3}{6}\right) + \left(\frac{2 * 3}{5 + 4}\right)}{2} = 0.58$$

According to HSMD measure, we have $R_1 = 5+7 = 12$, $R_2 = 5$, and $G = (2 + 5 + 7 + 8 + 9) * (9 + 6 + 5 + 1) = 651$. Hence, HSMD measure is calculated according to equation 14.2, as follows:

$$\text{HSMD} = 1 - \frac{12 * 5 + 1}{651} = 0.91$$

When HSMD measure is combined with Jaccard measure, it is called HSMDJ which is specified by equation 14.3.

$$\text{HSMDJ}(u_1, u_2) = \text{HSMD}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \quad (14.3)$$

Another extension of SMD called Amer measure is developed by Amer and Nguyen. Amer is more general than SMD because it concerns rating values with the aspect of relevant values and non-relevant values. Let r_m be median of rating values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$, hence, rating values which are greater or equal than r_m are called relevant ratings and the others which are smaller than r_m are called non-relevant. Let $\text{inc1}(u_1, u_2, j)$ and $\text{inc2}(u_1, u_2, j)$ be two increment functions on user 1, user 2, and item j which is defined as follows:

$$\text{inc1}(u_1, u_2, j) = \begin{cases} 1 & \text{if both } r_{1j} \text{ and } r_{2j} \text{ are non-missing} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{inc2}(u_1, u_2, j) = \begin{cases} 1 & \text{if both } r_{1j} \text{ and } r_{2j} \text{ are relevant} \\ 1 & \text{if both } r_{1j} \text{ and } r_{2j} \text{ are non-relevant} \\ 0 & \text{otherwise} \end{cases}$$

Essentially, Amer is derived from SMD by replacing the intersection cardinality $|I_1 \cap I_1|$ by sum of $\text{inc1}(u_1, u_2, j)$ and $\text{inc2}(u_1, u_2, j)$ over j . We define:

$$A = \frac{1}{2} \sum_{j \in I_1 \cup I_2} (\text{inc1}(u_1, u_2, j) + \text{inc2}(u_1, u_2, j))$$

The factor $\frac{1}{2}$ occurs in A because the sum of $\text{inc1}(u_1, u_2, j)$ and $\text{inc2}(u_1, u_2, j)$ aims to fold double the cooccurrence of two rated items. Therefore, Amer measure is defined as follows:

$$\text{Amer}(u_1, u_2) = \left(\frac{0.5}{|I_1 \cup I_2|} + \frac{1}{|I_1| + |I_2|} \right) A$$

In other words, Amer is specified more succinctly as follows:

$$\begin{aligned} \text{Amer}(u_1, u_2) &= \left(\frac{0.5}{|I_1| + |I_2|} + \frac{0.25}{|I_1 \cup I_2|} \right) \\ &\quad * \sum_{j \in I_1 \cup I_2} (\text{inc1}(u_1, u_2, j) + \text{inc2}(u_1, u_2, j)) \end{aligned} \quad (14.4)$$

For example, given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$, according to Amer measure, we have:

$$\begin{aligned} \text{Amer}(u_1, u_2) &= \left(\frac{0.5}{5+4} + \frac{0.25}{9} \right) * \sum_{j \in I_1 \cup I_2} (\text{inc1}(u_1, u_2, j) + \text{inc2}(u_1, u_2, j)) \\ &= \frac{0.75}{9} \sum_{j \in I_1 \cup I_2} (\text{inc1}(u_1, u_2, j) + \text{inc2}(u_1, u_2, j)) \end{aligned}$$

We also have $\text{inc1}(u_1, u_2, 1) = 1, \text{inc2}(u_1, u_2, 1) = 0, \text{inc1}(u_1, u_2, 2) = 0, \text{inc2}(u_1, u_2, 2) = 0, \text{inc1}(u_1, u_2, 3) = 1, \text{inc2}(u_1, u_2, 3) = 0, \text{inc1}(u_1, u_2, 4) = 1, \text{inc2}(u_1, u_2, 4) = 1, \text{inc1}(u_1, u_2, 5) = 0, \text{inc2}(u_1, u_2, 5) = 0, \text{inc1}(u_1, u_2, 6) = 1, \text{inc2}(u_1, u_2, 6) = 0$. Therefore,

$$\begin{aligned} \text{Amer}(u_1, u_2) &= \left(\frac{0.5}{5+4} + \frac{0.25}{9} \right) * \sum_{j \in I_1 \cup I_2} (\text{inc1}(u_1, u_2, j) + \text{inc2}(u_1, u_2, j)) \\ &= \frac{0.75}{9} ((1+0) + (0+0) + (1+0) + (1+1) + (0+0) + (1+0)) \\ &= \frac{0.75 * 5}{9} \cong 0.42 \end{aligned}$$

We develop a new measure which is based on HSMD measure and ideology of TF and IDF. Firstly, we research deeply HSMD measure in which the ratio $R_1 R_2 / G$ indicates difference between u_1 and u_2 . In other words, such ratio implies uniqueness of each vector, which means that the ratio $R_1 R_2 / G$ follows ideology of document frequency (DF) in information retrieval. Hence, essentially HSMD measure is a *quasi-IDF*. Here we re-define the quasi-IDF as follows:

$$\text{quasiIDF}(u_1, u_2) = 1 - \frac{(\sum_{j \in I_1 \setminus I_2} r_{1j})(\sum_{j \in I_2 \setminus I_1} r_{2j})}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})} \quad (14.5)$$

Note, notation “ $\setminus$ ” denotes complement operator in set theory. The quasi-IDF measure associated with Jaccard derives quasi-IDFJ measures as follows:

$$\text{quasiIDFJ}(u_1, u_2) = 1 - \frac{(\sum_{j \in I_1 \setminus I_2} r_{1j})(\sum_{j \in I_2 \setminus I_1} r_{2j}) * (1 - \text{Jaccard}(u_1, u_2))}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})} \quad (14.6)$$

Recall that Jaccard measure is mentioned in previous section. Similarly, following ideology of term frequency (TF), we define a so-called quasi-TF as follows:

$$\text{quasiTF}(u_1, u_2) = \frac{(\sum_{j \in I_1 \cap I_2} r_{1j})(\sum_{j \in I_1 \cap I_2} r_{2j})}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})} \quad (14.7)$$

Note, notation “ $\cap$ ” denotes intersection operator in set theory. The quasi-TF measure associated with Jaccard derives quasi-TFJ measures as follows:

$$\text{quasiTFJ}(u_1, u_2) = \frac{(\sum_{j \in I_1 \cap I_2} r_{1j})(\sum_{j \in I_1 \cap I_2} r_{2j}) * \text{Jaccard}(u_1, u_2)}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})} \quad (14.8)$$

The new measure called quasi-TF-IDF (QTI) is product of the quasi-TF and the quasi-IDF as follows:

$$\begin{aligned} \text{QTI}(u_1, u_2) &= \text{quasiTF}(u_1, u_2) * \text{quasiIDF}(u_1, u_2) \\ &= \frac{(\sum_{j \in I_1 \cap I_2} r_{1j})(\sum_{j \in I_1 \cap I_2} r_{2j})}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})} * \left(1 - \frac{(\sum_{j \in I_1 \setminus I_2} r_{1j})(\sum_{j \in I_2 \setminus I_1} r_{2j})}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})}\right) \end{aligned} \quad (14.9)$$

QTI is associated with Jaccard measure, which derives QTIJ measure as follows:

$$\begin{aligned} \text{QTIJ}(u_1, u_2) &= \text{quasiTFJ}(u_1, u_2) * \text{quasiIDFJ}(u_1, u_2) \\ &= \frac{(\sum_{j \in I_1 \cap I_2} r_{1j})(\sum_{j \in I_1 \cap I_2} r_{2j}) * \text{Jaccard}(u_1, u_2)}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})} \\ &\quad * \left(1 - \frac{(\sum_{j \in I_1 \setminus I_2} r_{1j})(\sum_{j \in I_2 \setminus I_1} r_{2j}) * (1 - \text{Jaccard}(u_1, u_2))}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})}\right) \end{aligned} \quad (14.10)$$

Given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$, we calculate QTIJ as an example. Of course, we have $I_1 = \{1, 2, 3, 4, 6\}$ and $I_2 = \{1, 4, 5, 6\}$. We also have:

$$\begin{aligned} \text{quasiTFJ}(u_1, u_2) &= \frac{(2 + 8 + 9)(9 + 6 + 1)}{(2 + 5 + 7 + 8 + 9)(9 + 6 + 5 + 1)} * \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} = \frac{19 * 16}{31 * 21} * \frac{3}{6} \cong 0.23 \\ \text{quasiIDFJ}(u_1, u_2) &= 1 - \frac{(5 + 7) * 5}{(2 + 5 + 7 + 8 + 9)(9 + 6 + 5 + 1)} * \left(1 - \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|}\right) \\ &= 1 - \frac{12 * 5}{31 * 21} \left(1 - \frac{3}{6}\right) \cong 0.95 \end{aligned}$$

This implies:

$$\text{QTIJ}(u_1, u_2) = \text{quasiTFJ}(u_1, u_2) * \text{quasiIDFJ}(u_1, u_2) = 0.23 * 0.95 \cong 0.22$$

EMX proposed by Amer is an improvement of Amer measure by combining Amer measure with other measures such as URP, NNSM, Coco, and STB. The integrations of Amer and URP, NNSM are called EMX1 and EMX2 defined as follows:

$$\text{EMX1}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2|)}\right) * \text{NNSM} * \text{Amer} \quad (14.11)$$

$$\text{EMX2}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2| |\sigma_1 - \sigma_2|)}\right) * \text{NNSM} * \text{Amer} \quad (14.12)$$

Where measures URP and NNSM are defined by equations 26.1 and 15.1. The first part in the equations above specifies URP. The integrations of Amer and URP, Coco are called EMX3 and EMX4 defined as follows:

$$\text{EMX3}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2|)}\right) * \text{Coco} * \text{Amer} \quad (14.13)$$

$$\text{EMX4}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2| |\sigma_1 - \sigma_2|)}\right) * \text{Coco} * \text{Amer} \quad (14.14)$$

Where Coco is defined by equation 25.1. The integrations of Amer and URP, STB are called EMX5 and EMX6 defined as follows:

$$\text{EMX5}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2|)}\right) * \text{STB} * \text{Amer} \quad (14.15)$$

$$\text{EMX6}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2| |\sigma_1 - \sigma_2|)}\right) * \text{STB} * \text{Amer} \quad (14.16)$$

Where STB is defined by equation 27.1. Equations of EMX measures are repeated in section 26.

15. NNSM

Jaccard measure, which is an effective similarity measures, focuses on whether items are rated but it does not concern magnitude rating values like other measures. We overcome this drawback by proposing a so-called numerical nearby similarity measure (NNSM) which concerns magnitude rating values and keeps strong point of Jaccard measure. In other words, NNSM combines sums of rating values and cardinalities of item sets. Equation 15.1 specifies NNSM.

$$\text{NNSM}(u_1, u_2) = \frac{|I_1 \cap I_2| \sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{|I_1| \sum_{j \in I_1} r_{1j} + |I_2| \sum_{j \in I_2} r_{2j}} \quad (15.1)$$

Note that $|I_1 \cap I_2|$ is the number of items rated by both user 1 and user 2, $|I_1|$ is the number of items rated by only user 1, and $|I_2|$ is the number of items rated by only user 2. It is easy to recognize that NNSM is an interesting advanced variant of cosine measure with support of Jaccard measure. However, NNSM is totally different from combination of cosine and Jaccard as CosineJ. Experimental section will mention evaluation of NNSM and CosineJ. Anyway, NNSM is simpler than CosineJ.

Given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$, we calculate NNSM as an example. Of course, we have $I_1 = \{1, 2, 3, 4, 6\}$ and $I_2 = \{1, 4, 5, 6\}$. We also have:

$$\text{NNSM}(u_1, u_2) = \frac{|I_1 \cap I_2|(2 * 9 + 8 * 6 + 9 * 1)}{|I_1|(2 + 5 + 7 + 8 + 9) + |I_2|(9 + 6 + 5 + 1)} = \frac{3 * 75}{5 * 31 + 4 * 21} \cong 0.94$$

NNSM is associated with Jaccard measure, which derives NNSMJ measure as follows:

$$\text{NNSMJ}(u_1, u_2) = \text{NNSM}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \quad (15.2)$$

16. TA

Cosine measure is effective but it has a drawback that there may be two end points of two vectors which are far from each other according to Euclidean distance, but their cosine is high. This is negative effect of Euclidean distance which decreases accuracy of cosine similarity. Therefore, a so-called triangle area (TA) measure (Nguyen & Amer, 2019) is proposed as an improved version of cosine measure. Figure 17.1 illustrates TA measure.

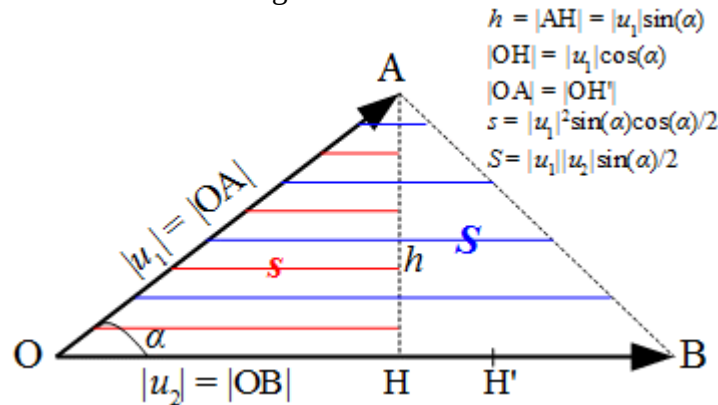


Figure 17.1. Triangle area (TA) measure with $0 \leq \alpha \leq \pi/2$

TA measure uses ratio of basic triangle area to whole triangle area as reinforced factor for Euclidean distance so that it can alleviate negative effect of Euclidean distance whereas it keeps simplicity and effectiveness of both cosine measure and Euclidean distance in making similarity of two vectors. TA is considered as an advanced cosine measure. TA is defined by equation 16.1 (Nguyen & Amer, 2019):

$$\begin{aligned}
u_1 \cdot u_2 \geq 0: \text{TA}(u_1, u_2) &= \begin{cases} \frac{(u_1 \cdot u_2)^2}{|u_1|(|u_2|)^3} & \text{if } |u_1| \leq |u_2| \\ \frac{(u_1 \cdot u_2)^2}{(|u_1|)^3|u_2|} & \text{if } |u_1| > |u_2| \end{cases} \\
u_1 \cdot u_2 < 0: \text{TA}(u_1, u_2) &= \begin{cases} \frac{u_1 \cdot u_2}{(|u_2|)^2} & \text{if } |u_1| \leq |u_2| \\ \frac{u_1 \cdot u_2}{(|u_1|)^2} & \text{if } |u_1| > |u_2| \end{cases}
\end{aligned} \tag{16.1}$$

Where, as usual:

$$\begin{aligned}
u_1 \cdot u_2 &= \sum_{j \in I_1 \cap I_2} r_{1j} r_{2j} \\
|u_1| &= \sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j})^2} \\
|u_2| &= \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j})^2}
\end{aligned}$$

Let TAJ denote the combined measure which combines TA measure and Jaccard measure. TAJ measure is defined as follows:

$$\text{TAJ}(u_1, u_2) = \text{TA}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \tag{16.2}$$

Let r_m be median of rating values, TA measure is normalized as TAN measure as follows:

$$\begin{aligned}
\text{TAN}(u_1, u_2) &= \text{TA}(u_1, u_2) \\
u_1 \cdot u_2 &= \sum_{j \in I_1 \cap I_2} (r_{1j} - r_m)(r_{2j} - r_m) \\
|u_1| &= \sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - r_m)^2} \\
|u_2| &= \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - r_m)^2}
\end{aligned} \tag{16.3}$$

By combined with Jaccard measure, TAN measure becomes TANJ measure as follows:

$$\text{TANJ}(u_1, u_2) = \text{TAN}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \tag{16.4}$$

As a convention, TA family includes TA, TAJ, TAN, and TANJ. Hence, equation 16.1 is the key of TA family.

17. RA

Chen et al. (Chen, Zhang, Liu, Gao, & Zhou, 2016) consider the rating matrix as a user-item (user-object) bipartite network in which every link in the network represents the rating that a user rated on an item. Chen et al. (Chen, Zhang, Liu, Gao, & Zhou, 2016, p. 608) stated that “the resource-allocation (RA) process is equivalent to the one-step random walk in the user-object bipartite network starting from the common neighbors”. Therefore, the RA index between user 1 and user 2 is (Chen, Zhang, Liu, Gao, & Zhou, 2016, p. 608):

$$\text{RA}(u_1, u_2) = \sum_{j \in I_1 \cap I_2} \frac{r_{1j} r_{2j}}{|v_j|^2} \tag{17.1}$$

Where $|v_j|$ is module of item j ,

$$|v_j| = \sqrt{\sum_{i \in J_j} r_{ij}^2}$$

Recall that J_j is the set of users who rated on item j . In the original article, Chen et al. actually calculated the RA index between two items, but their model can be extended to both item-based NN algorithm and user-based NN algorithm. Chen et al. (Chen, Zhang, Liu, Gao, & Zhou, 2016, pp. 608 - 609) combine cosine measure and RA index to derive a new measure called CosRA as follows:

$$\text{CosRA}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} \frac{r_{1j} r_{2j}}{|v_j|^2}}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j})^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j})^2}} \quad (17.2)$$

It is interesting to extend CosRA measure of Chen et al. as a combination of RA index and Pearson measure as follows:

$$\text{PearsonRA}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} \frac{(r_{1j} - \bar{u}_1)(r_{2j} - \bar{u}_2)}{|v_j|^2}}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - \bar{u}_2)^2}} \quad (17.3)$$

18. Entropy

Lee (Lee, Entropy-weighted similarity measures for collaborative recommender systems, 2018) used entropy to improve similarity measure; exactly Lee attached information entropy to measures such as cosine and Pearson. Let r_{min} and r_{max} be the minimum value and maximum value of ratings and let $r(j)$ denote the rating for item j (users who rated such item j are not concerned yet), the entropy of item j denoted $E(j)$ is defined as follows (Lee, Improving Jaccard Index for Measuring Similarity in Collaborative Filtering, 2017, p. 3):

$$E(j) = - \sum_{v=r_{min}}^{r_{max}} P(r(j) = v) \log_2(P(r(j) = v)) \quad (18.1)$$

Note, $P(r(j)=v)$ is the probability that users rated item j with rating value v (Lee, Improving Jaccard Index for Measuring Similarity in Collaborative Filtering, 2017, p. 3).

$$P(r(j) = v) = \frac{|u \in \mathbf{U}: r_{uj} = v|}{|u \in \mathbf{U}: r_{uj} \text{ not missing}|}$$

Note, r_{uj} is the rating that user u rated on item j . Finally, Lee (Lee, Improving Jaccard Index for Measuring Similarity in Collaborative Filtering, 2017, p. 3) developed the entropy cosine and entropy Pearson by attaching the entropy to cosine and Pearson as follows:

$$\text{cos}_e(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j} E(j)}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j})^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j})^2}} \quad (18.2)$$

$$\text{pearson}_e(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)(r_{2j} - \bar{u}_2) E(j)}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j} - \bar{u}_1)^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j} - \bar{u}_2)^2}} \quad (18.3)$$

19. Kullback - Leibler divergence

The drawback of traditional measures such as cosine and Pearson is that they are calculated based on the common items that two concerned users rated. Therefore, Deng et al. (Deng, et

al., 2018, p. 571) proposed the similarity model which does not require such common rated items, as follows:

$$\text{sim}(u_1, u_2) = \sum_{j \in I_1} \sum_{k \in I_2} \text{sim}_{\text{item}}(j, k) * s(r_{1j}, r_{2k}) \quad (19.1)$$

The quantity $s(r_{1j}, r_{2k})$ is called local similarity function based on sigmoid function, which is defined as follows (Deng, et al., 2018, p. 571):

$$s(r_{1j}, r_{2k}) = \frac{1}{1 + \exp(-(r_{1j} - \bar{u}_1)(r_{2k} - \bar{u}_2))} \quad (19.2)$$

Where $\bar{u}_1$ and $\bar{u}_2$ are mean values of u_1 and u_2 , respectively.

$$\bar{u}_1 = \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \bar{u}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j}$$

The quantity $\text{sim}_{\text{item}}(j, k)$ is the similarity measure of item j and item k which does not require common (co-rating) users. Deng et al. applied Kullback - Leibler (KL) divergence into calculating the similarity $\text{sim}_{\text{item}}(j, k)$ as follows (Deng, et al., 2018, p. 572):

$$\text{sim}_{\text{item}}(j, k) = \sum_{v=r_{\min}}^{r_{\max}} p_{jv} \log \frac{p_{jv}}{p_{kv}} \quad (19.3)$$

Note, $r_{\min}$ and $r_{\max}$ are the minimum value and maximum value of ratings, respectively. Where p_{jv} (p_{kv}) is the probability of rating value v that item j (item k) received among its ratings (Deng, et al., 2018, p. 572).

$$p_{jv} = \frac{|u \in \mathbf{U}: r_{uj} = v|}{|u \in \mathbf{U}: r_{uj} \text{ not missing}|}$$

$$p_{kv} = \frac{|u \in \mathbf{U}: r_{uk} = v|}{|u \in \mathbf{U}: r_{uk} \text{ not missing}|}$$

Therefore, the measure model of Deng et al. is also called KL measure. Note, KL divergence is a concept in information theory like entropy.

20. OS

Gazdar and Hidri (Gazdar & Hidri, 2019) proposed a so-called Our developed Similarity (OS) measure when taking account kernel function. Their OS is product of the so-called Percentage of Non-Common Ratings (PNCR) and the Absolute Difference of Ratings (ADR), as follows (Gazdar & Hidri, 2019, p. 19):

$$\text{OS}(u_1, u_2) = \text{PNCR}(u_1, u_2) * \text{ADR}(u_1, u_2) \quad (20.1)$$

Where (Gazdar & Hidri, 2019, p. 19),

$$\text{PNCR}(u_1, u_2) = \exp\left(-\frac{|\mathbf{V}| - |I_1 \cap I_2|}{|\mathbf{V}|}\right) \quad (20.2)$$

$$\text{ADR}(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} \exp\left(-\frac{|r_{1j} - r_{2j}|}{\max(r_{1j}, r_{2j})}\right)}{|I_1 \cap I_2|} \quad (20.3)$$

Note, $|\mathbf{V}|$ is the number of all items and $\max(r_{1j}, r_{2j})$ is the maximum value among r_{1j} and r_{2j} . Moreover, quantities $\text{PNCR}(u_1, u_2)$ and $\text{ADR}(u_1, u_2)$ can be considered as independent similarity measures.

21. SMC

Verma and Aggarwal (Verma & Aggarwal, 2019) proposed the Simple Matching Coefficient (SMC) measure by using statistical concept “statistical ratio match”. Let r_m be median of rating values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$, two rating values r_{1j} and r_{2j} match together if there are three cases as follows (Verma & Aggarwal, 2019, p. 42):

1. They are equal as $r_{1j} = r_{2j}$.
2. Both of them are smaller than r_m as $r_{1j} < r_m$ and $r_{2j} < r_m$.
3. Both of them are larger than r_m as $r_{1j} > r_m$ and $r_{2j} > r_m$.

Therefore, Verma and Aggarwal defined SMC as the ratio of the number of matched items to the total number of items as follows (Verma & Aggarwal, 2019, p. 42):

$$SMC(u_1, u_2) = \frac{|\{j \in I_1 \cap I_2 : r_{1j} \text{ matches } r_{2j}\}|}{|I_1 \cap I_2|} \quad (21.1)$$

22. ESim

ESim measure developed by Ali Amer and implemented by Loc Nguyen alleviates incomplete problem by ignoring missing values instead of aligning non-missing values of both rating vectors. Moreover, ESim removes second-order operators from its equation. In particular, ESim is defined as follows:

$$ESim(u_1, u_2) = |I_1 \cap I_2| \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{\sum_{j \in I_1} r_{1j} + \sum_{j \in I_2} r_{2j}} \quad (22.1)$$

It is easy to recognize that ESim, which is similar to NNSM, is a variant of cosine measure with support of Jaccard measure except that ESim does not concern the length of rating vector when it focuses on the additional module of rating vectors. Recall that the equation of NNSM is:

$$NNSM(u_1, u_2) = \frac{|I_1 \cap I_2| \sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{|I_1| \sum_{j \in I_1} r_{1j} + |I_2| \sum_{j \in I_2} r_{2j}}$$

For example, given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$, we calculate ESim as follows:

$$ESim(u_1, u_2) = \frac{|I_1 \cap I_2| (2 * 9 + 8 * 6 + 9 * 1)}{(2 + 5 + 7 + 8 + 9) + (9 + 6 + 5 + 1)} = \frac{3 * 75}{31 + 21} \cong 4.33$$

ZSim is a variant of ESim in which its denominator is replaced by product of rating values, as follows:

$$ZSim(u_1, u_2) = |I_1 \cap I_2| \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})} \quad (22.2)$$

It is easy to recognize that ZSim is more similar to Cosine than ESim. For example, given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$, we calculate ZSim as follows:

$$ZSim(u_1, u_2) = \frac{|I_1 \cap I_2| (2 * 9 + 8 * 6 + 9 * 1)}{(2 + 5 + 7 + 8 + 9) * (9 + 6 + 5 + 1)} = \frac{3 * 75}{31 * 21} \cong 0.35$$

ESim2 is a variant of ZSim where its nominator is improved by ignoring missing values as follows:

$$ESim2(u_1, u_2) = |I_1 \cap I_2| \frac{\sum_{j \in I_1 \cap I_2} \alpha_{1j} \alpha_{2j}}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})} \quad (22.3)$$

Where,

$$\alpha_{1j} = \begin{cases} 1 & \text{if } r_{1j} \text{ missing and } r_{2j} \text{ non-missing} \\ r_{1j} & \text{otherwise} \end{cases}$$

$$\alpha_{2j} = \begin{cases} 1 & \text{if } r_{2j} \text{ missing and } r_{1j} \text{ non-missing} \\ r_{2j} & \text{otherwise} \end{cases}$$

For example, given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$, we calculate ESIM2 as follows:

$$\text{ESim2}(u_1, u_2) = |I_1 \cap I_2| * \frac{(2 * 9 + 5 * 1 + 7 * 1 + 8 * 6 + 1 * 5 + 9 * 1)}{(2 + 5 + 7 + 8 + 9) * (9 + 6 + 5 + 1)} = \frac{3 * 92}{31 * 21} \cong 0.42$$

Both ZSim and ESIM2 are strengthened with the number of commonly rating values $|I_1 \cap I_2|$. ESIM3 is another variant of ZSim, which is strengthened by the number of non-missing values for both user 1 and user 2, as follows:

$$\text{ESim3}(u_1, u_2) = |I_1 \cup I_2| \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{(\sum_{j \in I_1} r_{1j})(\sum_{j \in I_2} r_{2j})} \quad (22.4)$$

For example, given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$, we calculate ZSim as follows:

$$\text{ESim3}(u_1, u_2) = \frac{|I_1 \cup I_2|(2 * 9 + 8 * 6 + 9 * 1)}{(2 + 5 + 7 + 8 + 9) * (9 + 6 + 5 + 1)} = \frac{6 * 75}{31 * 21} \cong 0.69$$

Measure IZSM developed is a variant of ZSim measure, in which missing values are replaced by 1 in case that rating values are greater than 1.

$$\text{IZSM}(u_1, u_2) = |I_1 \cap I_2| \frac{\sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{(\sum_{j \in I_1} (r_{1j} + \beta_{1j}))(\sum_{j \in I_2} (r_{2j} + \beta_{2j}))}$$

Where,

$$\beta_{1j} = \begin{cases} 1 & \text{if } r_{1j} \text{ missing} \\ 0 & \text{otherwise} \end{cases}$$

$$\beta_{2j} = \begin{cases} 1 & \text{if } r_{2j} \text{ missing} \\ 0 & \text{otherwise} \end{cases}$$

IZSM will be only effective if rating values are larger than 1. For example, given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$, we calculate IZSM as follows:

$$\text{ZSim}(u_1, u_2) = |I_1 \cap I_2| * \frac{(2 * 9 + 8 * 6 + 9 * 1)}{(2 + 5 + 7 + 8 + 1 + 9) * (9 + 1 + 1 + 6 + 5 + 1)} = \frac{3 * 75}{32 * 23} \cong 0.31$$

ESim is modified so as to derive a so-called Nearby measure as follows:

$$\text{Nearby}(u_1, u_2) = \frac{|I_1 \cap I_2| \sum_{j \in I_1 \cap I_2} r_{1j} r_{2j}}{|I_1| \sum_{j \in I_1} r_{1j} + |I_2| \sum_{j \in I_2} r_{2j}}$$

Indeed, Nearby measure is NNSM measure.

23. RES

By inspiring the physical resonance of simple harmonic motion, Tan and He (Tan & He, 2017) proposed a new similarity resonance similarity (RES) measure. The believed that user rating behavior is similar to the simple harmonic motion of particles in vibration system. RES is product of three components such as consistence component, distance component, and Jaccard component.

$$\text{RES}(u_1, u_2) = \left(\sum_{j \in I_1 \cap I_2} C(u_1, u_2, v_j, k_1) D(u_1, u_2, v_j, k_2, k_3) \right) J(u_1, u_2, k_4) \quad (23.1)$$

Note, v_j denotes item j whereas k_1, k_2, k_3 , and k_4 are predefined constants. Let $r_{\min}$ and $r_{\max}$ be the minimum value and maximum value of ratings, respectively and let r_m be median of rating

values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$. Let μ be the mean over all rating values and let $\bar{u}_i$ and $\bar{v}_j$ be rating means of user i and item j .

$$\bar{u}_i = \frac{1}{|I_i|} \sum_{j \in I_i} r_{ij}$$

$$\bar{v}_j = \frac{1}{|J_j|} \sum_{i \in J_j} r_{ij}$$

Consistency component $C(u_1, u_2, v_j, k_1)$ reflecting the harmonic motion is calculated based on the initial phase φ of harmonic vibration $x(t) = A \cos(\omega t + \varphi)$, as follows:

$$C(u_1, u_2, v_j, k_1) = \left(0.5 + 0.5 \cos(\varphi_{1j} - \varphi_{2j}) \right)^{\frac{k_1}{2}} \quad (23.2)$$

Where,

$$\varphi_{ij} = \begin{cases} \frac{\pi}{r_{max} - r_{min}} \varphi_{ij}^+ & \text{if } (r_{ij} - r_m)(\bar{u}_i - \mu) \geq 0 \\ \frac{\pi}{r_{max} - r_{min}} \varphi_{ij}^- & \text{if } (r_{ij} - r_m)(\bar{u}_i - \mu) < 0 \end{cases}$$

Note,

$$\varphi_{ij}^+ = \frac{r_{ij} - r_m}{1 + |\bar{u}_i - \mu|}$$

$$\varphi_{ij}^- = \left(1 + \frac{|\bar{u}_i - \mu|}{(r_{max} - r_{min})/2} \right) (r_{ij} - r_m)$$

Distance component reflecting distance between two ratings is defined as follows:

$$D(u_1, u_2, v_j, k_2, k_3) = \begin{cases} D^+(u_1, u_2, v_j, k_2) & \text{if } (r_{1j} - \bar{v}_j)(r_{2j} - \bar{v}_j) \geq 0 \\ D^-(u_1, u_2, v_j, k_3) & \text{if } (r_{1j} - \bar{v}_j)(r_{2j} - \bar{v}_j) < 0 \end{cases} \quad (23.3)$$

Where,

$$D^+(u_1, u_2, v_j, k_2) = \exp \left(-|r_{1j} - r_{2j}| + \frac{1}{2} (|r_{1j} - \bar{v}_j| + |r_{2j} - \bar{v}_j|) \right)^{k_2}$$

$$D^-(u_1, u_2, v_j, k_3) = \exp(-|r_{1j} - r_{2j}|)^{k_3}$$

Jaccard component is Jaccard measure powered with constant k_4 .

$$J(u_1, u_2, k_4) = \left(\frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} \right)^{k_4} \quad (23.4)$$

24. SM

Singularity implies an item that very few users rated it and so it is often ignored in calculating similarity but such item is valuable to recognizing two similar users. For instance, if both users rated the same singular items, they are highly similar together. Bobadilla et al. (Bobadilla, Ortega, & Hernando, 2012) propose a so-called singularity measure (SM) to concern singular items. Let r_m be median of rating values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$, hence, rating values which are greater or equal than r_m are called relevant ratings and the others which are smaller than r_m are called non-relevant. Let R_j and N_j are sets of users who gave relevant ratings and non-relevant ratings on item j , respectively. Thus, S_R^j is defined as the singularity of item j with regard to relevant values and S_N^j is defined as the singularity of item j with regard to non-relevant values.

$$\begin{aligned}
S_R^j &= 1 - \frac{|R_j|}{|U|} \\
S_N^j &= 1 - \frac{|N_j|}{|U|}
\end{aligned} \tag{24.1}$$

Essentially, SM is an extension of the mean squared difference (MSD) measure with support of singularity quantities like S_R^j and S_N^j , which is defined as follows:

$$\begin{aligned}
SM(u_1, u_2) &= \frac{1}{3} \left(\frac{1}{|A|} \sum_{j \in A} (1 - (r_{1j} - r_{2j})^2) (S_R^j)^2 \right. \\
&\quad + \frac{1}{|B|} \sum_{j \in B} (1 - (r_{1j} - r_{2j})^2) (S_N^j)^2 \\
&\quad \left. + \frac{1}{|C|} \sum_{j \in C} (1 - (r_{1j} - r_{2j})^2) S_R^j S_N^j \right)
\end{aligned} \tag{24.2}$$

Where A is the set of items which are received relevant ratings from both user 1 and user 2, B is the set of items which are received non-relevant ratings from both user 1 and user 2, and C is the set of items which are received relevant ratings from one user and non-relevant ratings from the other.

25. Coco

Cosine measure is based on dot product which is to make multiplication on pairs of elements obtained from two rating vectors. Because cosine measure only concerns commonly rated items, gaps between two vectors can be enlarged when there are a lot of elements inside the two vectors but there are few common elements. Derived from ideology of dot product, a so-called circle product aims to filling spaces inside pairs of vector elements by taking multiplication over all vector elements, as follows:

$$u_1 \otimes u_2 = \sum_{j \in I_1} \sum_{k \in I_2} r_{1j} r_{2k} \tag{25.1}$$

From the equation above, it is easy recognize that circle product is only valid if elemental dimensions have the same meaning. In the context of rating matrix, all rating values have the same meaning as ratings that users gave on items within the same scale. Indeed, circle product is the product of sum of two vectors.

$$u_1 \otimes u_2 = \sum_{j \in I_1} \sum_{k \in I_2} r_{1j} r_{2k} = \left(\sum_{j \in I_1} r_{1j} \right) \left(\sum_{k \in I_2} r_{2k} \right)$$

The so-called circle length is defined based on circle product, as follows:

$$|u_1|_{\otimes} = \sqrt{|u_1 \otimes u_2|} = \left| \sum_{j \in I_1} r_{1j} \right| \tag{25.2}$$

In this research, a so-called Coco measure is proposed based on circle product and normal vector length as follows:

$$\begin{aligned}
\text{Coco}(u_1, u_2) &= \frac{u_1 \otimes u_2}{|u_1||u_2|} = \frac{\sum_{j \in I_1} \sum_{k \in I_2} r_{1j} r_{2k}}{\sqrt{\sum_{j \in I_1 \cap I_2} (r_{1j})^2} \sqrt{\sum_{j \in I_1 \cap I_2} (r_{2j})^2}} \\
&= \frac{(\sum_{j \in I_1} r_{1j})(\sum_{k \in I_2} r_{2k})}{\sqrt{(\sum_{j \in I_1 \cap I_2} (r_{1j})^2)(\sum_{j \in I_1 \cap I_2} (r_{2j})^2)}}
\end{aligned} \tag{25.3}$$

In other words, Coco measure replaces dot product by circle product, which is its different aspect from cosine. For example, given two rating vectors $u_1 = (r_{11}=2, r_{12}=5, r_{13}=7, r_{14}=8, r_{15}=?, r_{16}=9)$ and $u_2 = (r_{21}=9, r_{22}=?, r_{23}=?, r_{24}=6, r_{25}=5, r_{26}=1)$, their Coco measure is:

$$\text{Coco}(u_1, u_2) = \frac{(2 + 5 + 7 + 8 + 9)(9 + 6 + 5 + 1)}{\sqrt{(2^2 + 5^2 + 7^2 + 8^2 + 9^2)(9^2 + 6^2 + 5^2 + 1^2)}} \cong 3.65$$

Coco associated with Jaccard produces CocoJ measure as follows:

$$\begin{aligned}
\text{CocoJ}(u_1, u_2) &= \text{Coco}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \\
&= \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} \frac{(\sum_{j \in I_1} r_{1j})(\sum_{k \in I_2} r_{2k})}{\sqrt{(\sum_{j \in I_1 \cap I_2} (r_{1j})^2)(\sum_{j \in I_1 \cap I_2} (r_{2j})^2)}}
\end{aligned} \tag{25.4}$$

26. URP

User rating preference (URP) measure is defined based on difference between two user ratings regarding their means and standard deviation as follows:

$$\text{URP}(u_1, u_2) = 1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2| |\sigma_1 - \sigma_2|)} \tag{26.1}$$

Where μ_1 and μ_2 are rating means of user 1 and user 2, respectively and σ_1 and σ_2 are rating standard deviations of user 1 and user 2, respectively.

$$\begin{aligned}
\mu_1 &= \frac{1}{|I_1|} \sum_{j \in I_1} r_{1j}, \mu_2 = \frac{1}{|I_2|} \sum_{j \in I_2} r_{2j} \\
\sigma_1 &= \sqrt{\frac{1}{|I_1|} \sum_{j \in I_1} (r_{1j} - \mu_1)^2}, \sigma_2 = \sqrt{\frac{1}{|I_2|} \sum_{j \in I_2} (r_{2j} - \mu_2)^2}
\end{aligned}$$

EMX proposed by Amer is an improvement of URP by combining URP with other measures such as Amer, NNSM, Coco, and STB. The integrations of URP and Amer, NNSM are called EMX1 and EMX2 defined as follows:

$$\text{EMX1}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2|)}\right) * \text{NNSM} * \text{Amer} \tag{26.2}$$

$$\text{EMX2}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2| |\sigma_1 - \sigma_2|)}\right) * \text{NNSM} * \text{Amer} \tag{26.3}$$

Where measures NNSM and Amer are defined by equations 15.1 and 14.4. The integrations of URP and Coco, Amer are called EMX3 and EMX4 defined as follows:

$$\text{EMX3}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2|)}\right) * \text{Coco} * \text{Amer} \tag{26.4}$$

$$\text{EMX4}(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2| |\sigma_1 - \sigma_2|)}\right) * \text{Coco} * \text{Amer} \tag{26.5}$$

Where Coco is defined by equation 25.1. The integrations of URP and STB, Amer are called EMX5 and EMX6 defined as follows:

$$EMX5(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2|)}\right) * STB * Amer \quad (26.5)$$

$$EMX6(u_1, u_2) = \left(1 - \frac{1}{1 + \exp(-|\mu_1 - \mu_2||\sigma_1 - \sigma_2|)}\right) * STB * Amer \quad (26.7)$$

Where STB is defined by equation 27.1.

27. STB

Amer proposed a so-called STB measure which is focused on product of sum of rating values called product-sum. In particular, STB is similarity of two rating vectors. Let X be product-sum of common part:

$$X = \left(\sum_{j \in I_1 \cap I_2} r_{1j} \right) \left(\sum_{j \in I_1 \cap I_2} r_{2j} \right)$$

Let Y be product-sum of different part:

$$Y = \left(\sum_{j \in I_1 \setminus I_2} r_{1j} \right) \left(\sum_{j \in I_2 \setminus I_1} r_{2j} \right)$$

Note, notation “ $\setminus$ ” denote complement operator in set theory. Let Z be the whole product-sum:

$$Z = \left(\sum_{j \in I_1} r_{1j} \right) \left(\sum_{j \in I_2} r_{2j} \right)$$

STB is similarity of two rating vectors with regard to product-sums, defined as follows:

$$STB = \frac{X}{Z} \left(1 - \frac{Y}{Z} \right)$$

28. RPB

RPB measure is trigonometric cosine function of statistical deviations, specified as follows (Ayub M. , et al., 2019, p. 6):

$$RPB(u_1, u_2) = \cos(|\bar{u}_1 - \bar{u}_2| |\text{var}(u_1) - \text{var}(u_2)|) \quad (28.1)$$

Note, $\bar{u}_i$ and $\text{var}(u_i)$ are mean and variance of u_i , respectively.

$$\bar{u}_i = \frac{1}{|I_i|} \sum_{j \in I_i} r_{ij}, \text{var}(u_i) = \frac{1}{|I_i|} \sum_{j \in I_i} (r_{ij} - \bar{u}_i)^2$$

In practice, the variance of u_i can be replaced by the standard deviation of u_i as follows (Ayub M. , et al., 2019, p. 6):

$$RPB(u_1, u_2) = \cos(|\bar{u}_1 - \bar{u}_2| |\text{sd}(u_1) - \text{sd}(u_2)|) \\ \text{sd}(u_i) = \sqrt{\text{var}(u_i)}$$

29. ADR

Absolute Difference of Ratings (ADR) is defined as follows (Gazdar & Hidri, 2019, p. 19),

$$ADR(u_1, u_2) = \frac{\sum_{j \in I_1 \cap I_2} \exp\left(-\frac{|r_{1j} - r_{2j}|}{\max(r_{1j}, r_{2j})}\right)}{|I_1 \cap I_2|} \quad (29.1)$$

Note, $\max(r_{1j}, r_{2j})$ is the maximum value among r_{1j} and r_{2j} .

30. Singularity and CLAG

Jin et al. (Jin, Zhang, Cai, & Zhang, 2020) combined local similarity and global similarity with support of singularity factor. As a result, they proposed a so-called Combined Local and Global (CLAG) measure. Let r_m be median of rating values, for example, if rating values range from 1 to 5, the median is $r_m = (1+5) / 2 = 3$, hence, rating values which are greater or equal than r_m are called relevant ratings and the others which are smaller than r_m are called non-relevant. Jin et al. called “relevant” and “non-relevant” as “positive” and “negative”, respectively. Given user 1 and user 2, for every item j , let:

- If both r_{1j} and r_{2j} are positive, item j belongs to the predefined set PA .
- If both r_{1j} and r_{2j} are negative, item j belongs to the predefined set NA .
- If r_{1j} and r_{2j} are different which means that one is positive and the other is negative, item j belongs to the predefined set D .

Let S_j^P and S_j^N be positive singularity and negative singularity

$$S_j^P = 1 - \frac{|P_j|}{|U|}$$

$$S_j^N = 1 - \frac{|N_j|}{|U|}$$

Where P_j is the set of users who gave positive ratings to item j and N_j is the set of users who gave negative ratings to item j . Jin et al. (Jin, Zhang, Cai, & Zhang, 2020, p. 6) defined their singularity measure as follows:

$$\text{singularity}(u_1, u_2) = \frac{1}{3} (L_P(u_1, u_2) + L_N(u_1, u_2) + L_D(u_1, u_2)) \quad (30.1)$$

Where,

$$L_P(u_1, u_2) = \frac{1}{|PA|} \sum_{j \in PA} \left(1 - \frac{1}{1 + \exp(-|r_{1j} - r_{2j}|)} \right) (S_j^P)^2$$

$$L_N(u_1, u_2) = \frac{1}{|NA|} \sum_{j \in NA} \left(1 - \frac{1}{1 + \exp(-|r_{1j} - r_{2j}|)} \right) (S_j^N)^2$$

$$L_D(u_1, u_2) = \frac{1}{|D|} \sum_{j \in D} \left(1 - \frac{1}{1 + \exp(-|r_{1j} - \bar{u}_1||r_{2j} - \bar{u}_2|)} \right) S_j^P S_j^N$$

Note, $\bar{u}_i$ is mean of u_i .

$$\bar{u}_i = \frac{1}{|I_i|} \sum_{j \in I_i} r_{ij}$$

According to Jin et al. (Jin, Zhang, Cai, & Zhang, 2020, p. 6), global measure is defined based on Jaccard measure as follows:

$$\text{global}(u_1, u_2) = 1 - \exp \left(- \frac{|I_1 \cap I_2|}{|I_1 \cup I_2|} \right) \quad (30.2)$$

Let $\{b_1, b_2, \dots, b_n\}$ be bins of rating values, for instance, if rating values range from 1 to 5, these bins are $\{1, 2, 3, 4, 5\}$. Let $F_i = \{f_{i1}, f_{i2}, \dots, f_{in}\}$ be the frequency vector of user i with regard to the bins $\{b_1, b_2, \dots, b_n\}$. Concretely, f_{ij} is the number of items that user i rated with bin b_j . Jin et al. (Jin, Zhang, Cai, & Zhang, 2020, p. 6) defined the bin measure which is proportioned to dot product of F_1 and F_2 regarding user 1 and user 2.

$$\text{bin}(u_1, u_2) = \sum_{j=b_1}^{b_n} \sqrt{f_{1j} f_{2j}} \quad (30.3)$$

Finally, Jin et al. (Jin, Zhang, Cai, & Zhang, 2020, p. 6) proposed their CLAG measure as the product of singularity measure, global measure and bin measure as follows:

$$\text{CLAG}(u_1, u_2) = \text{singularity}(u_1, u_2) * \text{global}(u_1, u_2) * \text{bin}(u_1, u_2) \quad (30.4)$$

31. EXP

Exponential measure are the measures that take advantages of exponential function which can be considered as sigmoid function. They are denoted as EXP measure. Particularly, EXP takes exponential function of MSD measure, which is smooth technique to alleviate small errors in evaluating differences between two vectors. Moghadam et al. (Moghadam, Heidari, Moeini, & Kamandi, 2019) proposed their **EXP** as follows:

$$\text{EXP}(u_1, u_2) = \exp \left(- \frac{\sum_{j \in I} (r_{1j} - r_{2j})^2}{(r_{\max} - r_{\min}) \sqrt{|I_1 \cup I_2|}} \right) \quad (31.1)$$

Where $r_{\min}$ and $r_{\max}$ are minimum rating value and maximum rating value, respectively, for example, if rating values range from 1 to 5, then $r_{\min} = 1$ and $r_{\max} = 5$. The quantity $|I_1 \cup I_2|$ indicates the number of items rated by both users.

Appendix B

Because this report only lists basic, main, or important measures whereas there are a huge number of variants and combinations which derived from measure groups, this appendix B lists some derived measures in set B from the paired label XY.

Replacing the built-in Jaccard2 in NHMS by SMD and Amer produce measures called NHMS.SMD and NHMS.Amer, respectively as follows:

$$\text{NHMS.SMD}(u_1, u_2) = \text{PSS}(u_1, u_2) * \text{URP}(u_1, u_2) * \text{SMD}(u_1, u_2) \quad (\text{B7.1})$$

$$\text{NHMS.Amer}(u_1, u_2) = \text{PSS}(u_1, u_2) * \text{URP}(u_1, u_2) * \text{Amer}(u_1, u_2) \quad (\text{B7.2})$$

Where PSS, URP, SMD, and Amer are specified by equations 7.1, 26.1, 14.1, and 14.4, respectively.

SMD associated with Jaccard produce a so-called SMDJ measure as follows:

$$\text{SMDJ}(u_1, u_2) = \text{SMD}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \quad (\text{B14.1})$$

Where SMD and Jaccard are specified by equations 14.1 and 1.1, respectively.

Amer associated with Jaccard produces measures called AmerJ as follows:

$$\text{AmerJ}(u_1, u_2) = \text{Amer}(u_1, u_2) * \text{Jaccard}(u_1, u_2) \quad (\text{B14.2})$$

Appendix C

Because this report only lists basic, main, or important measures whereas there are a huge number of variants and combinations which derived from measure groups, this appendix C lists some derived measures in set C from the paired label XY.

IJ associated with cosine, Pearson, PSS, Amer, TA, and Coco produces measures called IJ.Cosine, IJ.Pearson, IJ.PSS, IJ.Amer, IJ.TA, and IJ.Coco, respectively as follows:

$$\text{IJ.Cosine}(u_1, u_2) = \text{IJ}(u_1, u_2) * \text{cosine}(u_1, u_2) \quad (\text{C1.1})$$

$$\text{IJ.Pearson}(u_1, u_2) = \text{IJ}(u_1, u_2) * \text{Pearson}(u_1, u_2) \quad (\text{C1.2})$$

$$\text{IJ.PSS}(u_1, u_2) = \text{IJ}(u_1, u_2) * \text{PSS}(u_1, u_2) \quad (\text{C1.3})$$

$$\text{IJ.Amer}(u_1, u_2) = \text{IJ}(u_1, u_2) * \text{Amer}(u_1, u_2) \quad (\text{C1.4})$$

$$\text{IJ.TA}(u_1, u_2) = \text{IJ}(u_1, u_2) * \text{TA}(u_1, u_2) \quad (\text{C1.5})$$

$$IJ.Coco(u_1, u_2) = IJ(u_1, u_2) * Coco(u_1, u_2) \quad (C1.6)$$

Where IJ, cosine, Pearson, PSS, Amer, TA, and Coco are specified by equations 1.4, 2.1, 7.1, 14.4, 16.1, and 25.3, respectively.

Amer associated with cosine, Pearson, CPC, PSS, QTI, NNSM, and TA produces measures called Amer.Cosine, Amer.Pearson, Amer.CPC, Amer.PSS, Amer.QTI, Amer.NNSM, and Amer.TA, respectively as follows:

$$Amer.Cosine(u_1, u_2) = Amer(u_1, u_2) * cosine(u_1, u_2) \quad (C14.1)$$

$$Amer.Pearson(u_1, u_2) = Amer(u_1, u_2) * Pearson(u_1, u_2) \quad (C14.2)$$

$$Amer.CPC(u_1, u_2) = Amer(u_1, u_2) * CPC(u_1, u_2) \quad (C14.3)$$

$$Amer.PSS(u_1, u_2) = Amer(u_1, u_2) * PSS(u_1, u_2) \quad (C14.4)$$

$$Amer.QTI(u_1, u_2) = Amer(u_1, u_2) * QTI(u_1, u_2) \quad (C14.5)$$

$$Amer.NNSM(u_1, u_2) = Amer(u_1, u_2) * NNSM(u_1, u_2) \quad (C14.6)$$

$$Amer.TA(u_1, u_2) = Amer(u_1, u_2) * TA(u_1, u_2) \quad (C14.7)$$

Where Amer, cosine, Pearson, CPC, PSS, QTI, NNSM, and TA are specified by equations 14.4, 2.1, 3.1, 3.2, 7.1, 14.5, 15.1, 16.1, respectively.

Appendix E

This appendix lists some examples. Here some examples related to Coco measures including Coco, CocoJ, and CocoIJ are described. For example, given two rating vectors $u_3 = (r_{31}=4, r_{32}=1, r_{33}=5, r_{34}=5)$, and $u_4 = (r_{41}=1, r_{42}=2, r_{43}=?, r_{44}=?)$ from table 0.1:

	Item 1	Item 2	Item 3	Item 4
User 1	$r_{11} = 1$	$r_{12} = 2$	$r_{13} = 1$	$r_{14} = 5$
User 2	$r_{21} = 2$	$r_{22} = 1$	$r_{23} = 2$	$r_{24} = 4$
User 3	$r_{31} = 4$	$r_{32} = 1$	$r_{33} = 5$	$r_{34} = 5$
User 4	$r_{41} = 1$	$r_{42} = 2$	$r_{43} = ?$	$r_{44} = ?$

About Jaccard measure, we have $I_3 = \{1, 2, 3, 4\}$ and $I_4 = \{1, 2\}$ and so we obtain from equation 1.1:

$$Jaccard(u_3, u_4) = \frac{|I_3 \cap I_4|}{|I_3 \cup I_4|} = \frac{2}{4} = 0.5$$

About IJ measure, we have $PA = \{\}$, $NA = \{1, 2\}$, $D = \{\}$, $PO = \{3, 4\}$, $NO = \{\}$, $P_1=1$, $N_1=3$, $E_1=0$, $P_2=0$, $N_2=4$, $E_2=0$, $P_3=1$, $N_3=2$, $E_3=1$, $P_4=3$, $N_4=0$, and $E_4=1$ when $r_m=3$. We also obtain $S_1^P = 1 - 1/4 = 3/4$, $S_1^N = 1 - 3/4 = 1/4$, $S_1^E = 1 - 0/4 = 1$, $S_2^P = 1 - 0/4 = 1$, $S_2^N = 1 - 4/4 = 0$, $S_2^E = 1 - 0/4 = 1$, $S_3^P = 1 - 1/4 = 3/4$, $S_3^N = 1 - 2/4 = 1/2$, $S_3^E = 1 - 1/4 = 3/4$, $S_4^P = 1 - 3/4 = 1/4$, $S_4^N = 1 - 0/4 = 1$, and $S_4^E = 1 - 1/4 = 3/4$. From equation 1.4, IJ measure of u_3 and u_4 is:

$$IJ(u_3, u_4) = \frac{S_1^N + S_2^N}{S_1^N + S_2^N + \sqrt{S_3^P S_3^E} + \sqrt{S_4^P S_4^E}} = \frac{1/4 + 0}{1/4 + 0 + \sqrt{3/4 * 3/4} + \sqrt{1/4 * 3/4}} \cong 0.1745$$

From equation 25.3, Coco measure of u_3 and u_4 is:

$$Coco(u_1, u_2) = \frac{(4 + 1 + 5 + 5)(1 + 2 + 0 + 0)}{\sqrt{(4^2 + 1^2 + 5^2 + 5^2)(1^2 + 2^2 + 0^2 + 0^2)}} \cong 2.46$$

From equation 25.4, CocoJ measure of u_3 and u_4 is:

$$CocoJ(u_1, u_2) = Coco(u_1, u_2) * Jaccard(u_1, u_2) \cong 0.5 * 2.465 \cong 1.23$$

From equation C1.6, the combined measure of Coco and IJ on u_3 and u_4 is:

$$IJ.Coco = IJ(u_1, u_2) * Coco(u_1, u_2) \cong 0.1745 * 2.46 \cong 0.43$$

References

- Al-Shamri, M. H. (2021, October 30). Similarity modifiers for enhancing the recommender system performance. *Applied Intelligence*. doi:10.1007/s10489-021-02900-7
- Ayub, M., Ghazanfar, A. M., Khan, T., & Saleem, A. (2020, May 12). An Effective Model for Jaccard Coefficient to Increase the Performance of Collaborative Filtering. (B. M. Ali, Ed.) *Arabian Journal for Science and Engineering*, 45(12), 9997 - 10017. doi:10.1007/s13369-020-04568-6
- Ayub, M., Ghazanfar, M. A., Mehmood, Z., Saba, T., Alharbey, R., Munshi, A. M., & Alrige, M. A. (2019, August 1). Modeling user rating preference behavior to improve the performance of the collaborative filtering based recommender systems. *PLoS ONE*, 14(8), e0220129. doi:10.1371/journal.pone.0220129
- Bag, S., Kumar, S. K., & Tiwari, M. K. (2019, January 11). An Effective Model for Jaccard Coefficient to Increase the Performance of Collaborative Filtering. (W. Pedrycz, Ed.) *Information Sciences*, 483(May 2019), 53 - 64. doi:10.1016/j.ins.2019.01.023
- Bobadilla, J., Ortega, F., & Hernando, A. (2012, March). A collaborative filtering similarity measure based on singularities. *Information Processing and Management*, 48(2), 204-217. doi:10.1016/j.ipm.2011.03.007
- Chen, L.-J., Zhang, Z.-K., Liu, J.-H., Gao, J., & Zhou, T. (2016, October 3). A vertex similarity index for better personalized recommendation. (H. W. Capel, Ed.) *Physica A*, 466(2017), 607 - 615. doi:10.1016/j.physa.2016.09.057
- Choi, K., & Suh, Y. (2013, January). A new similarity function for selecting neighbors for each target item in collaborative filtering. (H. Fujita, & J. Lu, Eds.) *Knowledge-Based Systems*, 37(2013), 146-153. doi:10.1016/j.knosys.2012.07.019
- Deng, J., Wang, Y., Guo, J., Deng, Y., Gao, J., & Park, Y. (2018, October 29). A similarity measure based on Kullback-Leibler divergence for collaborative filtering in sparse data. *Journal of Information Science*, 45(5), 656-675. doi:10.1177/0165551518808188
- Feng, J., Fengs, X., Zhang, N., & Peng, J. (2018, September 24). An improved collaborative filtering method based on similarity. (H. Wang, Ed.) *PLoS ONE*, 13(10), 1-18. doi:10.1371/journal.pone.0204003
- Gazdar, A., & Hidri, L. (2019, September 25). A new similarity measure for collaborative filtering based recommender systems. (J. Lu, Ed.) *Knowledge-Based Systems*, 188(2020). doi:10.1016/j.knosys.2019.105058
- Harris, E. F., & Sjøvold, T. (2018, September 7). Calculation of Smith's Mean Measure of Divergence for Intergroup Comparisons Using Nonmetric Data. (E. F. Harris, Ed.) *Dental Anthropology: A Publication of the Dental Anthropology Association*, 17(3), 83-93. doi:10.26575/daj.v17i3.152
- Hyung, J. A. (2008, January 2). A new similarity measure for collaborative filtering to alleviate the new user cold-starting problem. (W. Pedrycz, Ed.) *Information Sciences*, 178(1), 37-51. doi:10.1016/j.ins.2007.07.024
- Jin, Q., Zhang, Y., Cai, W., & Zhang, Y. (2020, January 10). A New Similarity Computing Model of Collaborative Filtering. *IEEE Access*, 8, 17594 - 17604. doi:10.1109/ACCESS.2020.2965595
- Jindal, A., Sharma, N., & Verma, V. (2022). Joyful Jaccard: An Analysis of Jaccard-Based Similarity Measures in Collaborative Recommendations. *International Conference on Artificial Intelligence and Sustainable Engineering: Select Proceedings of AISE 2020*. 1. Springer Nature. doi:10.1007/978-981-16-8542-2_3

- Lee, S. (2017). Improving Jaccard Index for Measuring Similarity in Collaborative Filtering. In K. Kim, & N. Joukov (Ed.), *Information Science and Applications 2017 (ICISA 2017)*. 424, pp. 799 - 806. Macau: Springer. doi:10.1007/978-981-10-4154-9_93
- Lee, S. (2018). Entropy-weighted similarity measures for collaborative recommender systems. *AIP Conference Proceedings*. 1982, pp. 1-6. AIP Publishing. doi:10.1063/1.5045417
- Liang, S., Ma, L., & Yuan, F. (2015, 10). A singularity-based user similarity measure for recommender systems. *International Journal of Innovative Computing Information and Control*, 11(5), 1629-1638. doi:10.24507/ijicic.11.05.1629
- Lin, Y.-S., Jiang, J.-Y., & Lee, S.-J. (2013, January 25). A Similarity Measure for Text Classification and Clustering. (J. Pei, Ed.) *IEEE Transactions on Knowledge and Data Engineering*, 26(7), 1575-1590. doi:10.1109/TKDE.2013.19
- Liu, H., Hu, Z., Mian, A., Tian, H., & Zhu, X. (2013, November 20). A new user similarity model to improve the accuracy of collaborative filtering. (H. Fujita, & J. Lu, Eds.) *Knowledge-Based Systems*, 56(2014), 156-166. doi:10.1016/j.knosys.2013.11.006
- Manochandar, S., & Punniyamoorthy, M. (2020, August 22). A new user similarity measure in a new prediction model for collaborative filtering. (M. Ali, & H. Fujita, Eds.) *Applied Intelligence*, 51(1), 586 - 615. doi:10.1007/s10489-020-01811-3
- Moghadam, P. H., Heidari, V., Moeini, A., & Kamandi, A. (2019, September 6). An exponential similarity measure for collaborative filtering. *SN Applied Sciences*, 1(10), 1-4. doi:10.1007/s42452-019-1142-8
- Mu, Y., Xiao, N., Tang, R., Luo, L., & Yin, X. (2019, January). An Efficient Similarity Measure for Collaborative Filtering. (R. Bie, Y. Sun, & J. Yu, Eds.) *Procedia Computer Science*, 147(2019), 416-421. doi:10.1016/j.procs.2019.01.258
- Nguyen, L., & Amer, A. A. (2019, October 17). Advanced Cosine Measures for Collaborative Filtering. (ITS, Ed.) *Adaptation and Personalization (ADP)*, 1(1), 21-41. doi:10.31058/j.adp.2019.11002
- Patra, B. K., Launonen, R., Ollikainen, V., & Nandi, S. (2015, March 1). A new similarity measure using Bhattacharyya coefficient for collaborative filtering in sparse data. (H. Fujita, & J. Lu, Eds.) *Knowledge-Based Systems*, 82, 163-177. doi:10.1016/j.knosys.2015.03.001
- Sarwar, B., Karypis, G., Konstan, J., & Riedl, J. (2001). Item-based Collaborative Filtering Recommendation Algorithms. *Proceedings of the 10th international conference on World Wide Web* (pp. 285-295). Hong Kong: ACM. doi:10.1145/371920.372071
- Singh, P. K., Sinha, S., & Choudhury, P. (2022, January 25). An improved item-based collaborative filtering using a modified Bhattacharyya coefficient and user-user similarity as weight. *Knowledge and Information Systems*, 64, 665-701. doi:10.1007/s10115-021-01651-8
- Sun, S.-B., Zhang, Z.-H., Dong, X.-L., Zhang, H.-R., Li, T.-J., Zhang, L., & Min, F. (2017, August 17). Integrating Triangle and Jaccard similarities for recommendation. (Q. Zou, Ed.) *PLoS ONE*, 12(8), 1-16. doi:10.1371/journal.pone.0183570
- Suryakant, & Mahara, T. (2016, January). A New Similarity Measure Based on Mean Measure of Divergence for Collaborative Filtering in Sparse Environment. (R. K. Venugopal, R. Buyya, & M. L. Patnaik, Eds.) *Procedia Computer Science*, 89(2016), 450-456. doi:10.1016/j.procs.2016.06.099
- Tan, Z., & He, L. (2017, November 29). An Efficient Similarity Measure for User-Based Collaborative Filtering Recommender Systems Inspired by the Physical Resonance Principle. *IEEE Access*, 5, 27211 - 27228. doi:10.1109/ACCESS.2017.2778424
- Torres Júnior, R. (2004). *Combining Collaborative and Content-based Filtering to Recommend Research Paper*. Universidade Federal do Rio Grande do Sul, Programa de Pós Graduação em Educação. Porto Alegre: Universidade Federal do Rio Grande do

Sul. Retrieved from
<http://www.lume.ufrgs.br/bitstream/handle/10183/5887/000432990.pdf;sequence=1>
Verma, V., & Aggarwal, R. K. (2019). A New Similarity Measure Based on Simple Matching Coefficient for Improving the Accuracy of Collaborative Recommendations. (M. He, & Y. V. Bodyanskiy, Eds.) *International Journal of Information Technology and Computer Science*, 11(6), 37-49. doi:10.5815/ijitcs.2019.06.05